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PALEOMAGNETISM OF THE MOENKOPI FORMATION, COLORADO PLATEAU: BASIS FOR LONG-TERM MODEL OF ACQUISITION OF CHEMICAL REMANENT MAGNETISM IN RED BEDS

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Abstract. Paleomagnetic studies of Moenkopi samples from several sites on the Colorado Plateau, including a complete 427-m section at Sinbad Valley, western Colorado, reveal the following: (1) the presence of both normal and reversed Triassic components in many of the same samples at many localities, (2) the close correlation of the magnetic polarity and intensity of magnetic remanence with the lithologic characteristics in the Sinbad Valley section, (3) a streaked pattern of the remanence directions in any one member in the Sinbad Valley section similar to that expected from polar wander, (4) a lack of progressive change from the base upward, and (5) the common intermediate remanence directions of weak intensity (suggesting the presence of two or more nearly balanced anti-parallel components) throughout the formation. The results of our investigation, combined with previously reported petrographic and paleomagnetic studies of the Moenkopi Formation, have led us to conclude that the principal remanence carried by the formation is chemical remanent magnetism (CRM) acquired diagenetically over a geologically long time interval. From the available data we have formulated the following model to explain how the CRM was acquired through natural processes acting intermittently at geologically reasonable rates. When initially deposited, the Moenkopi sediments consisted primarily of first-cycle detritus containing a few percent of iron-bearing accessory minerals such as biotite, hornblende, magnetite, and ilmenite. At some time after deposition, under favorable subsurface conditions, intrastratal alteration began, involving breakdown of iron-bearing silicate and oxide minerals, partial replacement of some iron-rich silicate grains by hematite, martitization of magnetite, growth of hematite pods and blebs in ilmenite and biotite, and precipitation of crystalline hematite in both primary and secondary voids. The authigenesis spanned at least two polarity events and probably continued over several tens of millions of years. Stratigraphic units with grossly similar lithologies underwent similar diagenetic alterations at rates and to extents that were different from other stratigraphic units having different lithologies. Because of variations in the rate and duration of alteration and variations in the length of normal versus reversed polarity events, any given lithologic unit tended to be dominated by one polarity. The model given above seems to be capable of explaining not only those magnetic and petrographic features found in the Moenkopi Formation but also those of most other red beds.

Introduction

In recent years a controversy has arisen over the interpretation of paleomagnetic data obtained from red beds. The nature of the controversy has been elaborated by Walker et al., [1981] and Turner [1980], and only the main points will be recapitulated here.

Many workers are convinced that red bed remanence faithfully mirrors the geomagnetic field present at (or soon after) the time of deposition. According to this view, remanence data from red beds can be used to elucidate both the large-scale as well as small-scale features of the ancient geomagnetic field. Within the framework of this view, at least three different ideas have been postulated to account for the rapid acquisition of remanence; namely, (1) depositional remanence resulting from physical alignment of detrital hematite grains [Elston and Purucker, 1979]; (2) chemical remanence, as a by-product of hematite authigenesis occurring within several thousand years after deposition [Helsley and Steiner, 1974; and Helsley, 1974]; (3) a combination of ideas 1 and 2 [Purucker et al., 1980].

Others, ourselves included, are of the opinion that none of the above explanations is viable and that red bed remanence primarily consists of multiple, commonly near-antiparallel components acquired through chemical alteration discontinuously over long time intervals. If this view is correct, the remanence in red beds rarely reflects the geomagnetic field at the time of deposition.

None of the models postulated to date has been capable of explaining the variation in remanence found in red beds. Particularly, it is problematical why some red beds possess consistent (apparently reliable) remanence, while other lithologically similar red beds do not.

In 1975 we began a detailed study of the paleomagnetic, rock magnetic, and petrographic properties of the Triassic Moenkopi Formation, a unit considered by many to be the most reliable red bed sequence for use in magnetostratigraphy. Our goal was to develop a geologically reasonable model that could explain the magnetic and petrographic properties in the Moenkopi Formation and which would be applicable to other red beds as well.

The results of our petrographic study have already been reported [Walker et al., 1981]. This paper provides most of the results of the rock magnetic and paleomagnetic study and, in light of the petrographic results and data

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available in the literature, presents a model of remanence acquisition for the Moenkopi based on common intrastratal alterations acting over geologically long time intervals.

Laboratory and Field Methods

Rock samples were taken either as cores, by means of a portable gasoline powered drill, or as hand specimens and oriented by means of a Brunton Compass. Hand specimens were taken only in a few cases (as indicated in the text) where the rock was too badly fractured, too weak, or inadequately exposed to enable successful drilling. To ensure obtaining fresh rock during hand sampling, the exposed surface was first stripped back and excavated to a depth of 0.25-0.50 m. Remanence measurements were made with a Schonstedt spinner magnetometer with a sensitivity of about 5×10^{-7} emu. To improve the signal-to-noise ratio, sample holders were made from essentially nonmagnetic glass-ceramic material.

The principal technique used in our paleomagnetic study was step-wise partial thermal demagnetization, carried out in a noninductively wound furnace contained in nested mu-metal shields. The field within the sample chamber was periodically checked and was never permitted to exceed 4 gammas. Maintenance of such a low field is essential when demagnetization temperatures approach the maximum blocking temperature of hematite grains. Otherwise, spurious components of TRM acquired during cooling can become as large or larger than the sought after residual moments.

Alternating field (af) demagnetization was not performed on our samples for several reasons. First, it had been shown that af demagnetization had little effect in reducing scatter of remanence directions in lithologically similar Moenkopi samples obtained only a few kilometers from Sinbad Valley [Helsley, 1969]. Second, we wished to compare our results with those of Helsley [1969], Helsley and Steiner [1974], and Lienert and Helsley [1980], and all of the polarity zonations reported by those workers were based largely, and in some cases entirely, on samples that had not been subjected to af demagnetization. Third, none of the samples collected in our study possessed both unusually high remanence intensities and anomalous directions--characteristics indicative of the presence of lightning-induced isothermal remanent magnetization (IRM)--for which af demagnetization is known to be an effective treatment. Our samples were largely unaffected by lightning primarily because all of our sampling sites were either in lightning-shadow zones (near the bases and middle parts of steep scarps) or in road cuts.

Thermomagnetic analysis, used in some cases as an aid in identifying magnetic phases, was performed on a sensitive electrobalance [Larson et al.,1975]. This technique is particularly useful in identifying small admixtures of relatively highly magnetic material (e.g., magnetite) amongst less magnetic phases such as hematite.

The sections of the Moenkopi Formation selected for study are located (1) in Colorado, in Sinbad Valley, about 18.6 km due southwest of Gateway, (2) in Arizona, near Gray Mountain, about 100 km north of Flagstaff, (3) in Utah, at Maverick Ridge about 5 km east of Natural Bridges National Monument, and (4) in Utah, in Cache Valley, about 4 km east of Arches National Park (Figure 1).

Samples also were taken from a conglomerate bed in the Permian Cutler Formation stratigraphically just below the Moenkopi, along the Dolores River, 3 km south of Gateway, Colorado.

Results

Most of our effort was directed toward investigation of the entire section of Moenkopi exposed along the eastern side of Sinbad Valley. At the other localities, study was restricted to limited parts of the available stratigraphic intervals.

Sinbad Valley, Colorado

The type magnetostratigraphic section for much of the Moenkopi Formation and the polarity zonation for much of the Triassic [Helsley, 1969; Helsley and Steiner, 1974] is based on the section exposed in the canyons of the Dolores River and its tributary, Salt Creek, south of Gateway, Colorado (see Figures 1 and 2).

We selected for study a section of the Moenkopi, about 427 m (1400 ft) thick, that crops out along the eastern side of Sinbad Valley, 6 to 9 km southwestward of the Dolores River (Figure 2). Because of the nature of the available exposures, it was necessary to sample the formation in two separate segments (designated sites 1A and 1B) to achieve complete coverage. Prominent marker beds which are conspicuous and laterally extensive, enabled us to tie the two sections together with no overlap or omission. In segment 1A we sampled the lower portion of the formation; at site 1B we collected samples from the upper part. We attempted to take samples at stratigraphic intervals of 1.5 m (5 ft) and were generally successful. In a few instances the rock was so

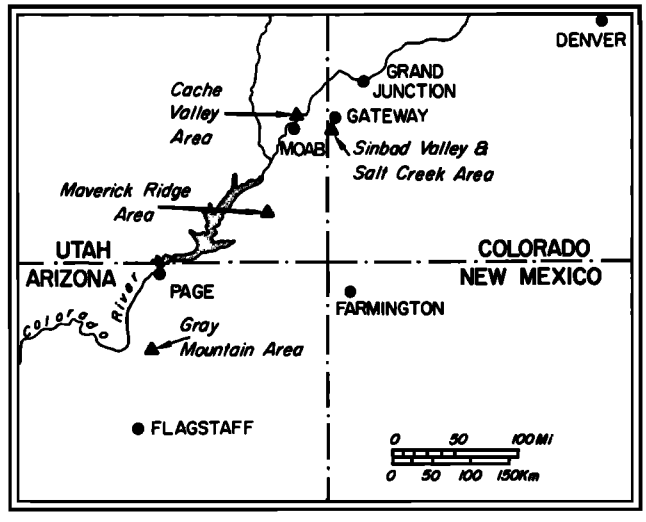


Fig. 1. Location map showing collecting sites.

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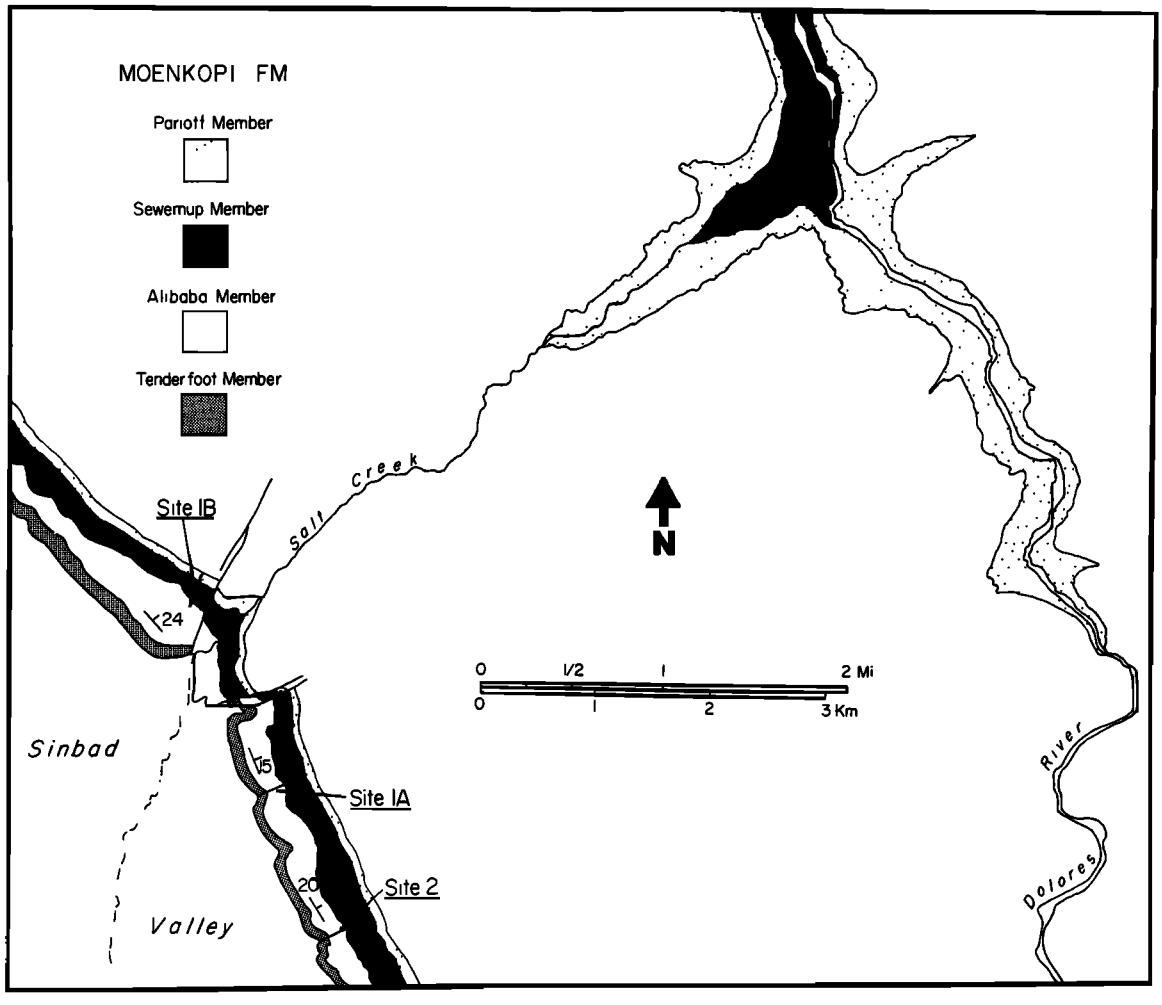


Fig. 2. Sketch map of the Sinbad Valley and Salt Creek region showing locations of sites 1A, 1B, and 2.

badly fractured that satisfactory samples could not be obtained, and in some cases, samples were taken at less than the 1.5 m interval.

We also sampled the lower portion of the formation at site 2, approximately 1.6 km south of site 1A (Fig. 2), in order to compare two separate paleomagnetic records from the same stratigraphic interval.

The stratigraphy and lithology of the Moenkopi section in Sinbad Valley have been given by Shoemaker [1955], Shoemaker and Newman [1959], and Cater and Craig [1970]. The designations of the four members described in this paper follow those defined by Shoemaker and Newman [1959]. From the base upward they are the Tenderfoot (poorly sorted muddy sandstone), Alibaba (interbedded sandstone, siltstone, and shale), Sewemup (mudstone and shale), and Pariott (interbedded sandstone, siltstone, and shale). Hand samples (rather than core samples) were taken only from the Pariott Member and the upper one-half of the Sewemup Member. In general, the Tenderfoot and Sewemup Members are poorly resistant slope formers whereas the Pariott and Alibaba Members hold up ridges and ledges (see Figure 3). The strata are primarily continental deposits composed of detritus derived from Precambrian rocks of the ancestral Uncompahgre uplands that were exposed only a few kilometers to the east during the latest Paleozoic to early Mesozoic. The exact age of the Moenkopi in Sinbad Valley is unknown. Shoemaker and Newman [1959], on the basis of stratigraphic correlation, consider it to be Triassic but acknowledge that the lower beds may be as old as Permian. Stewart et al., [1972] conclude that the section in Sinbad Valley (western Colorado) cannot be exactly correlated on the basis of lithology with other sections of the Moenkopi exposed elsewhere on the Colorado Plateau. They indicate that the Sinbad Valley strata occur stratigraphically below those exposed in northeastern Arizona, near Gray Mountain, the site of extensive paleomagnetic, rock magnetic and petrographic studies [Shoemaker and Purucker, 1974; Larson et al., 1976; Elston and Purucker, 1979; Bressler and Elston, 1980; Purucker et al., 1980; Walker et al., 1981].

Paleomagnetic analyses

Following NRM measurement, all samples were thermally demagnetized at 550 degrees C and

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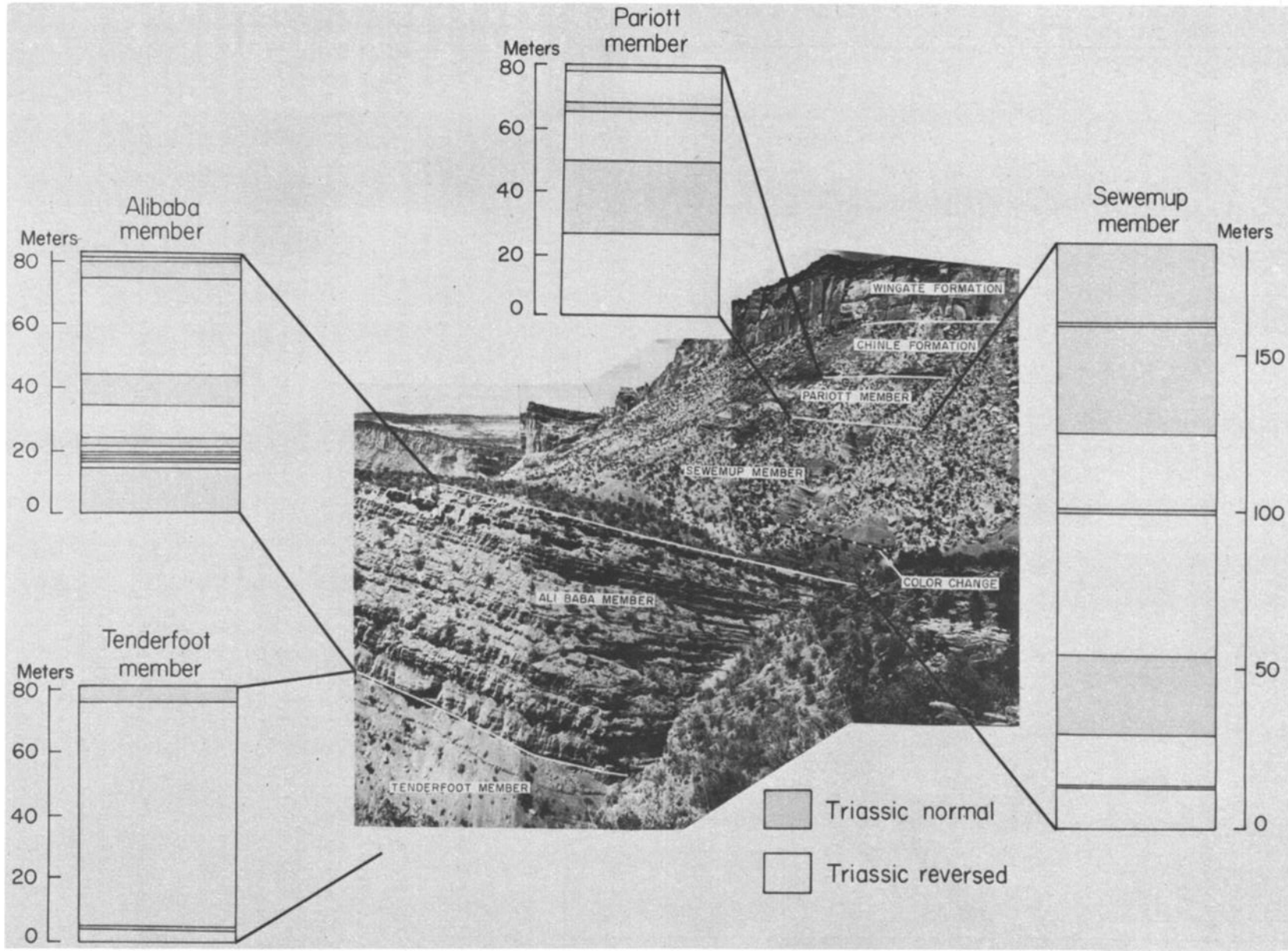


Fig. 3. Composite photograph of the Moenkopi Formation, taken near the base at site 1A, and the general remanence polarity zonation of each member (sites 1A and 1B) for the 550°C level of demagnetization.

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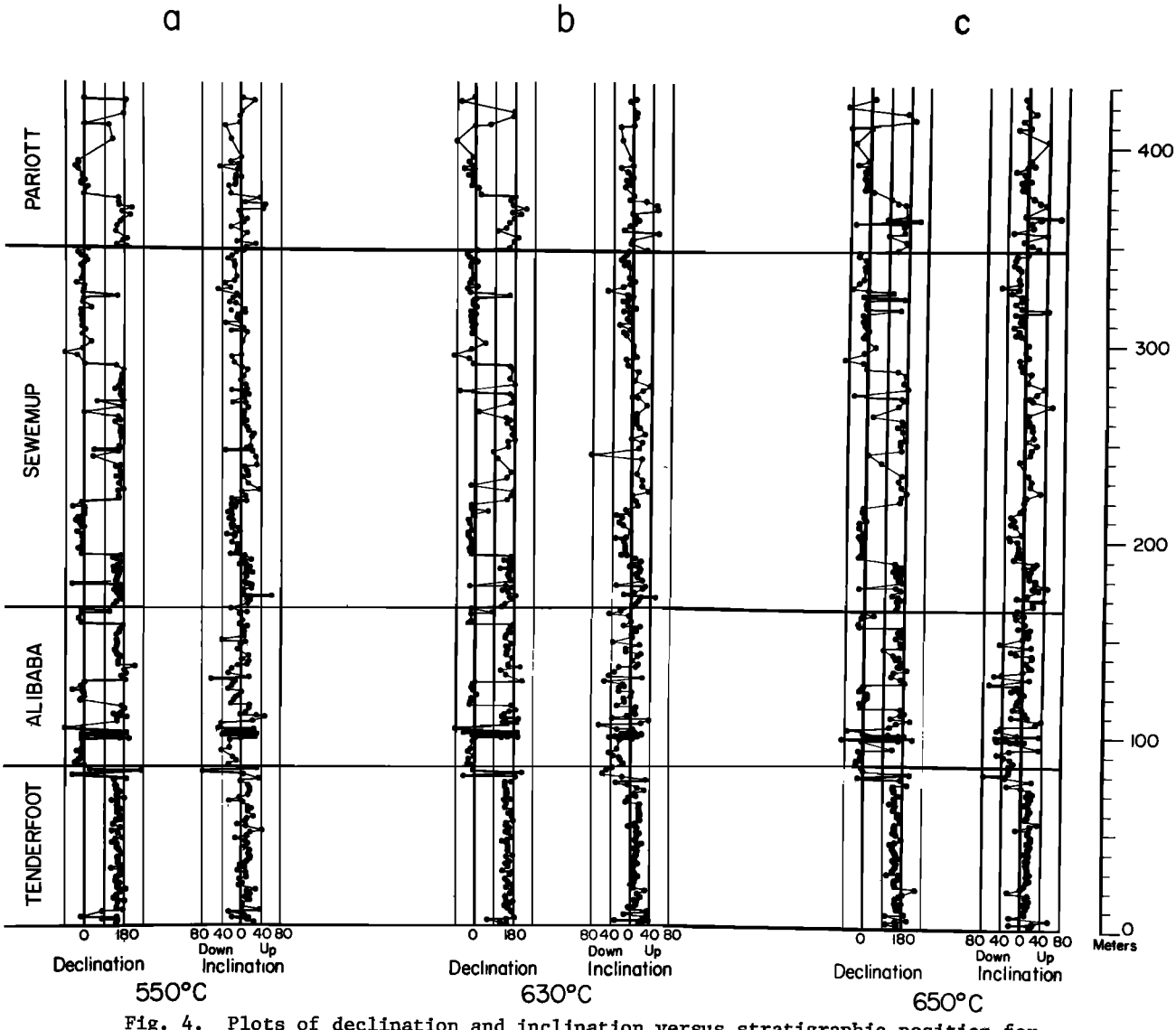


Fig. 4. Plots of declination and inclination versus stratigraphic position for combined sites 1A and 1B after demagnetization. (a) 550°C (b) 630°C (c) 650°C.

remeasured. Subsequently, most samples from sites 1A and 1B and selected samples from site 2 were demagnetized at 630°C, 650°C, and 670°C. The resulting data were analyzed in a number of ways, as discussed below.

Remanence Data of Individual Samples

Remanence directions (declination, inclination) versus stratigraphic position are given for site 1 samples after demagnetization at 550°C, 630°C, and 650°C (Figure 4) and for site 2 samples after demagnetization at 550°C (Fig. 5). To assess further the effect of demagnetization on a sample's remanence, we also calculated the vector removed during each progressive demagnetization step. Hereinafter, when discussing sample remanence directions or vectors lost during demagnetization the following notation will be used (see Figure 6 for pictorial representation). Normal (N) and reversed (R) remanence directions lie within 30° of the Triassic normal and reversed means, respectively, of the group of samples being analyzed. Any direction that is greater than 30° but less than 60° from the normal and reversed group mean is designated intermediate-normal (IN) and intermediate-reversed (IR), respectively. Those directions lying beyond the 60° limits, both of the normal and reversed group mean directions, are termed intermediate.

Sample number designations in the text correspond to stratigraphic elevations, in meters, above the base of the Moenkopi section.

Site 1

Tenderfoot Member

The remanence polarity pattern of the Tenderfoot Member, after demagnetization at 550°C, consists largely of a thick (76 m) lower, reversed Triassic section overlain by a thin (∿5 m) unit possessing intermediate-normal directions (Figure 4a). Sample 4.6 is exceptional because although it is in the reversed section, it displays a normal Triassic direction. Demagnetization at 630°C removed a

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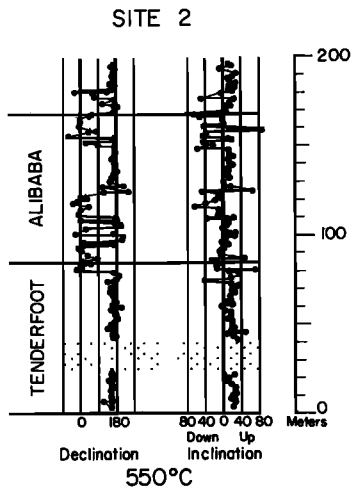


Fig. 5. Plot of declination and inclination versus stratigraphic position for site 2 after demagnetization at 550°C. That portion which lacks data (shaded) represents the zone where strata are missing due to faulting.

normal or intermediate-normal component from seven samples in the reversed section. After this cleaning step most samples in the lower unit display bona fide Triassic reversed directions; even the remanence in sample 4.6 moved to an intermediate-reversed direction (Figure 4b).

Subsequent demagnetization at 650°C produced little change in sample remanence polarities (Figure 4c). During this heating, however, N and IN components were lost from 11 samples in the reversed section, particularly in two stratigraphic intervals--one between 27.5 and 38 m and the other between 56.5 and 61 m. In the zone of intermediate remanence directions in the uppermost 5 m of the member, cleaning at 650°C resulted in loss of a reversed component from one of the four samples. After this step, two of the four samples in this zone display normal polarity, one is IN and one is intermediate. Demagnetization at 670°C led to an increase of scatter in directions throughout but to no large changes in general polarity zonation.

Alibaba Member

The remanent directions obtained from the lithologically diverse Alibaba Member are considerably more variable than those from the Tenderfoot Member (Figure 4), in that (1) there are many intermediate directions at all levels of thermal demagnetization, (2) many samples of one polarity lose components of the opposite polarity during the progressive demagnetization steps, and (3) progressive thermal demagnetization produces flip-flops in polarity in individual samples. For example, sample 98.3, which is from a reversely magnetized interval, exhibited a normal direction after demagnetization at 550°C; however, during demagnetization at 650°C it lost a normal component and became very nearly reversed. Sample 98.8, which after cleaning at 550°C displayed a normal direction, lost a reversed component in demagnetization at 630°C and changed to a reversed direction by 670°C; sample 111.3, which was reversely magnetized after demagnetization at 550°C, changed to nearly normal after the 630°C demagnetization step and after treatment at 650°C, lost a normal component and returned to a reversed direction.

Moreover, the stratigraphic polarity zonation at any given demagnetization step is complicated: Normal samples are present within sequences of reversed samples, and vice versa.

Sewemup Member

Compared to the Alibaba Member the uniformly fine-grained Sewemup Member is relatively simple in polarity zonation (Figure 4). After demagnetization at 550°C, the zonation is, from the base upward, reversed, normal, reversed, normal (Figure 4a). Even in this member, however, anomalous directions exist; samples with N or IN directions occur in reversed zones (samples 175.3 and 263.1) and vice versa (sample 323.2). Also, some samples changed either from normal to reversed or from reversed to normal during progressive thermal demagnetization. In some cases, the change occurred abruptly from one demagnetization level to the

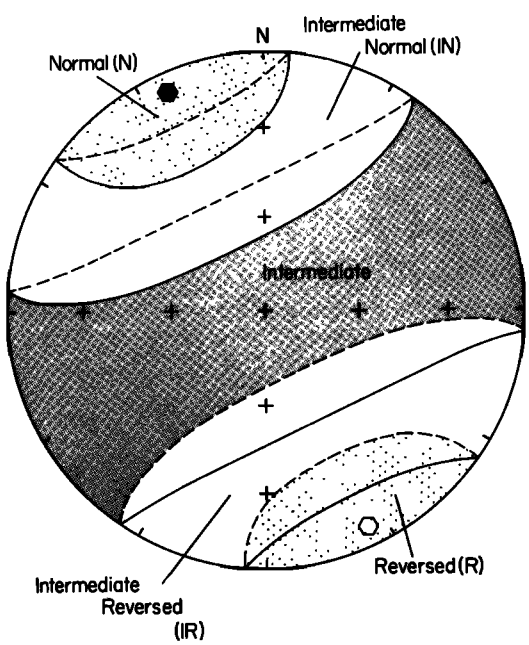


Fig. 6. Pictorial representation of terminology used in describing remanence vectors. Equal-area projection. Solid symbol and solid lines, lower hemisphere; open symbol and dashed lines, upper hemisphere. Hexagons represent mean normal and reversed Triassic remanence directions of a group of samples. Normal (N) and reversed (R) remanence directions lie within 30° of the normal and reversed means, respectively. Any direction beyond 30° but less than 60° from the normal and reversed means is designated intermediate-normal (IN) and intermediate-reversed (IR), respectively. Those directions lying beyond 60° from both the normal and reversed means, are termed intermediate.

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next (samples 191.2, 285.1, and 317.1); in others, it took place more or less progressively over several demagnetization steps (samples 274.7, 326.5, and 329). In addition, some samples changed not only from one polarity to the other but changed back again (samples 227 and 243.6).

For many samples, the vector erased at any one demagnetization level was essentially antiparallel to the sample direction before and/or after demagnetization.

Pariott Member

The polarity sequence of the Pariott Member, which, like the Alibaba Member, is composed of interbedded coarse- to fine-grained sandstone, siltstone, and shale, is somewhat complex (Figure 4). After demagnetization at 550°C samples from the lower part of the section were generally reversely magnetized, most of those from the middle portion were normally magnetized, and those in the upper part were of mixed polarity (Figure 4a). Of the 7 samples obtained from the 30-m-thick upper part, three are Triassic reversed, two are Triassic normal, one is IR and one is intermediate in direction. After demagnetization at 650°C it was apparent that the directions of some samples began to switch polarities. After the 670°C step, 7 out of 13 samples in the lower section became N, IN, or intermediate, whereas samples from the uppermost part of the section (402.1 to 421.6 m) were largely reversely magnetized. For many of the samples the vectors removed during successive demagnetization steps were largely antiparallel to sample remanence directions before and/or after demagnetization.

Site 2

All samples from this site, located about 1.6 km south of site 1A, were demagnetized at 550°C; the resulting polarity zonation is shown in Figure 5. Twelve samples were selected randomly and subjected to higher levels of cleaning.

Tenderfoot Member

After treatment at 550°C, the Tenderfoot Member can be divided into two polarity units: a very thick reversely magnetized lower zone, overlain by a thin normal to intermediate-normal zone (Figure 5). Within the reversed zone, about 13% of the samples possess intermediate or intermediate-reversed directions. Only two samples (44.7 and 58.4) were demagnetized to temperatures above 550°C. Between the 650° and 670° demagnetization step, sample 58.4 switched from a reversed to a nearly normal direction, whereas sample 44.7 remained reversely magnetized.

Alibaba Member

The polarity zonation in the Alibaba is very complex after demagnetization at 550°C (Figure 5). The polarity flip-flops between normal and reversed over short stratigraphic intervals,

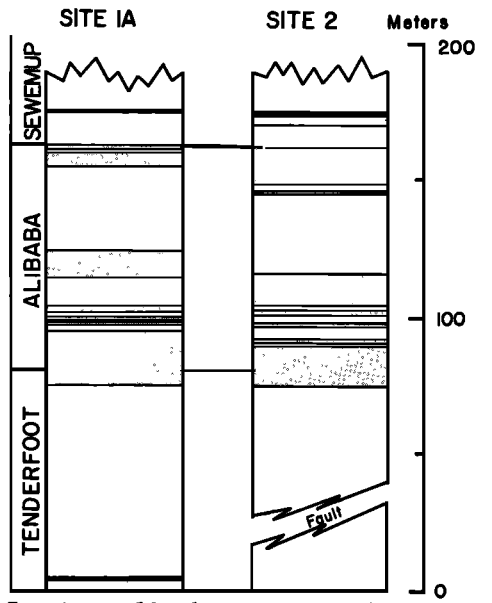


Fig. 7. Generalized remanence polarity zonations, after cleaning at 550°C, of the section sampled at site 2 and the equivalent stratigraphic section at site 1A. Sites separated by 1.6 km.

and samples possessing intermediate directions are common throughout the section.

Ten samples were demagnetized at temperatures above 550°C. Of these, four (nos. 79.5, 97.7, 102.3 and 126.7) switched polarity during the course of progressive demagnetization.

Sewemup Member

Only the basal 30 m of the Sewemup Member were sampled at site 2. After demagnetization at 550°C (Figure 5), the section is reversely magnetized except for a complicated zone within 10 m of the base that contains two thin nearly normally magnetized intervals. Only one sample (172.1) was cleaned to temperatures above 550°C, and it remained reversed in polarity throughout step-wise demagnetization.

Correlation of Sites 1A and 2

Correlation of the Tenderfoot, Alibaba, and lower part of the Sewemup members between sites 1A and 2, 1.6 km apart, can be made for the 550°C level of demagnetization (Figure 7). The polarity zonation in the Tenderfoot is very similar between the two sites although no normally magnetized samples are evident in the site 2 reversed section. In contrast, the match in polarity sequences for the Alibaba Member is poor. It is impossible to definitely correlate any one of the polarity zones or boundaries between the two sites. In the basal part of the Sewemup Member at site 1, only one thin normal zone is present, whereas at site 2, two normal zones are identifiable, separated by a thin reversed zone. This appears to be a real difference between the two sections, although it could also be the result of inadequate sampling density.

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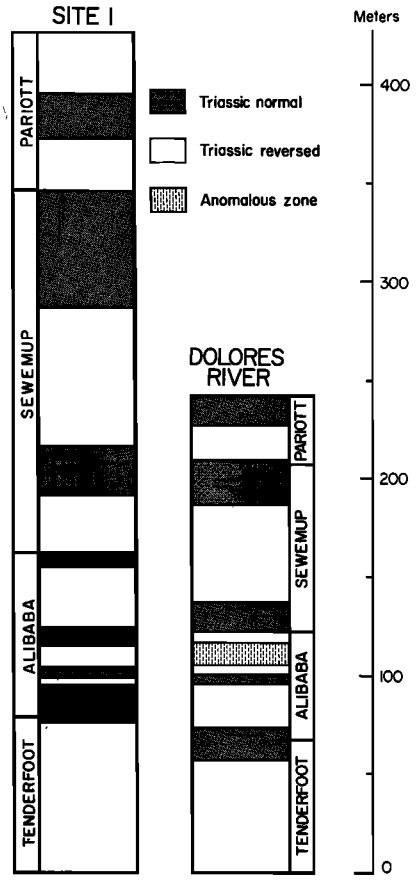


Fig. 8. Generalized magnetic polarity zonation of site 1, Sinbad Valley and Dolores River section [Helsley and Steiner, 1974]; in all cases, after demagnetization of samples at 550°C. In the Sinbad Valley zonation, single-sample intervals have been omitted in order to emphasize the general features. Note that in the Sinbad Valley section there is, comparatively, a greater number of polarity zones, a greater complexity of zonation in the Alibaba and Pariott members, and a lack of any anomalous zone in the Alibaba.

Evidence of Multiple Components

During progressive thermal demagnetization about 20% of the Moenkopi samples from Sinbad Valley undergo changes indicative of the presence of multiple antipodal components. The evidence includes both switches in sample polarity, and the occurrence of vectors removed at successive demagnetization levels which are nearly antiparallel to the sample direction. In order for antipodal vectors in a sample to become apparent, thermal treatment must bring about a change in their relative strengths. For samples in which the two components reside in hematitic materials with similar Curie- and blocking-temperature spectra, the relative magnitudes of the components generally will not be changed significantly by heating, and the presence of antipodal components will not be apparent. Therefore, it is likely that more than 20% of the samples contain multiple, antiparallel components.

Comparison of the Sinbad Valley and Dolores River Canyon Polarity Zonations

The Moenkopi section in Sinbad Valley is about twice as thick as that in the Dolores River Canyon, only 6 to 8 km away, where the magnetostratigraphy was first described by Helsley [1969] and Helsley and Steiner [1974]. This difference in thickness, coupled with variation in lithologies and lack of continuous exposures between the two sections, precludes making detailed stratigraphic correlation between the two areas. Accordingly, only general observations concerning comparison of the magnetic polarity zonation at the two locations can be made.

The generalized polarity sequences for the Sinbad Valley (site 1) and Dolores River sections, at the 550°C demagnetization level, are given in Figure 8. In the figure, all polarity zones based on only a single sample are excluded from the Sinbad Valley sequence. Comparison of the generalized sequences reveals that more polarity zones are present in the Sinbad Valley section than in the Dolores River section. Particularly, the sequence of zonation in both the Alibaba and Pariott members cannot be matched between the two areas. However, similarities do exist. For example, at both localities, the Tenderfoot is nearly entirely reversely magnetized, the Alibaba is composed of numerous relatively thin polarity zones, and the Sewemup, for its thickness, contains relatively few, thick zones.

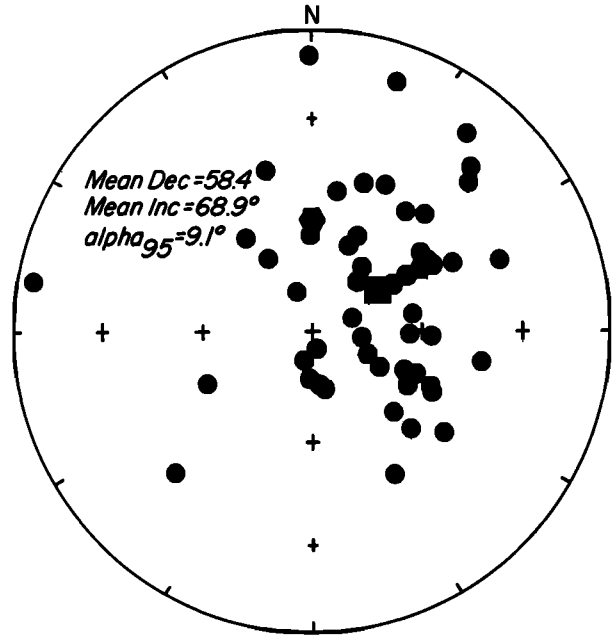


Fig. 9. Vectors removed during the initial demagnetization step at 550°C from samples of the Alibaba Member, site 1A, Sinbad Valley; equal-area projection, solid circles are downward. Present axial dipole field indicated by hexagon; mean direction by square. The mean of these vectors is essentially identical with that of samples in Helsley's [1969] and Helsley's and Steiner's [1974] anomalous zone in the Alibaba Member in the Dolores River Canyon.

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One obvious difference between the paleomagnetic records of the two areas is the absence of an anomalous zone in the Alibaba Member at Sinbad Valley, like that described in the Dolores River section (Figure 8). Helsley [1969] and Helsley and Steiner [1974] designated a zone about 11 m thick near the top of the Alibaba Member as an anomalous zone that displayed highly unusual remanence directions--mean declination, 50°, mean inclination, 70° downward--even after demagnetization at 600°C. Helsley and Steiner [1974] reported that the same zone could be found in two different sections separated by about 5 km and concluded that the zone must reflect anomalous behavior of the geomagnetic field during Triassic time.

The Alibaba Member in Sinbad Valley, although nearly 1.5 times as thick as the unit along the Dolores River, lacks any comparable zone of unusual magnetic directions. Indeed, a number of individual samples with intermediate directions are present, but they are located sporadically throughout the section. However, vectors removed during the first demagnetization step (550°C), from samples of the Alibaba at Sinbad Valley, form a cluster whose mean D (58.4°) and I (68.9°),is very similar in direction to that of the anomalous zone (Figure 9). In fact, demagnetization at 550°C removes similarly oriented components from essentially all samples of all members of the Moenkopi in Sinbad Valley. Based on their mean direction and relative low blocking temperatures (<550°C), it is likely that these components represent magnetization acquired relatively recently through near-surface chemical processes that produced very-fine-grained magnetic iron oxides. It is reasonable to conclude that the anomalous zone in the Dolores River section formed in the same way and bears little or no relation to anomalous Triassic geomagnetic field behavior.

Comparison of the Sinbad Valley/Dolores River Canyon Polarity Zonations with that at Maverick Ridge, Utah

Recently, the entire Moenkopi section at Maverick Ridge, 130 km southwest of Dolores River, was studied paleomagnetically and divided into geomagnetic polarity zones [Lienert and Helsley, 1980]. Sufficient variability exists in their data (see especially their Figure 6), that the zonation they report represents only one of several that could have been devised. In one case, a polarity zone (R2) is based on only two adjacent samples whereas in another, two adjacent normally magnetized samples in the reversed polarity zone R4 are dismissed as noise. Normally magnetized samples exist in reversely magnetized zones, and vice versa; samples with intermediate directions are common throughout the section (see their Figures 5 and 6). These features are very similar to those we have found in the Moenkopi section at Sinbad Valley and, we believe, are indicative of the presence of multiple, antiparallel components resulting from CRM acquisition. This conclusion is strongly supported by the results of detailed studies we made on two rock

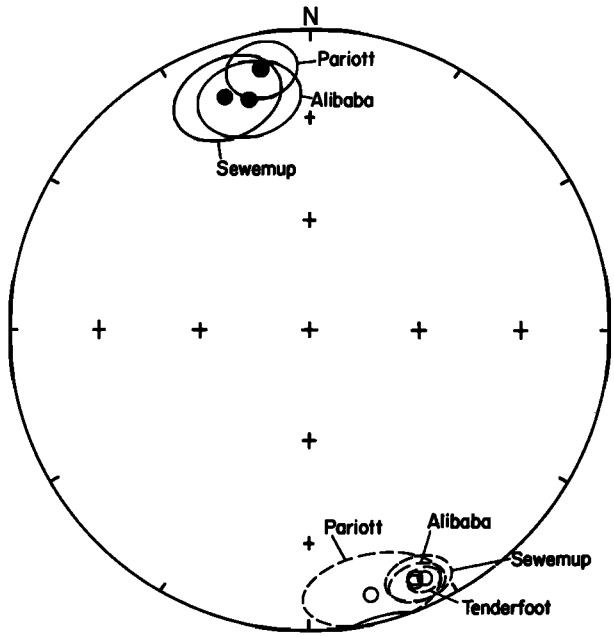


Fig. 10. Mean directions and 95% cones of confidence for the normal and reversed vectors of each member of the Moenkopi, Sinbad Valley, after demagnetization at 550°C: equal-area projection: solid circles, downward; open circles, upward. Because of lack of sufficient normal samples, the Tenderfoot mean was calculated only from the reversely magnetized population.

units at Maverick Ridge, which are given in a later section.

On the basis of minimal stratigraphic control, Lienert and Helsley [1980] correlated the section exposed at Maverick Ridge with the lower portion of the Moenkopi at Dolores River. Moreover, all polarity zones from N1 at the base, upward to R4, which were designated in the Dolores River section [Helsley and Steiner, 1974] were identified in the Maverick Ridge sequence. However, none of the supposedly correlatable unusual remanence features of the Dolores River section--such as the anomalous zone and cyclical variation in declination--are present in the Maverick Ridge section, and the relative thicknesses of the polarity units are completely dissimilar between the two areas.

In view of these difficulties, the fact that dissimilarities in the polarity zonation of the Moenkopi exist between the Dolores River section and the Sinbad Valley section we studied, over a distance of only 6 to 8 km, and previously mentioned problems involved in unambiguously defining the polarity zonation at Maverick Ridge, we are of the opinion that the magnetostratigraphic correlation between the Maverick Ridge and Dolores River sections is inadequately documented.

Relation of Polarity Zonation to Lithology

There is a strong correlation between lithology and both the character of the polarity zonation and presence of antiparallel components in the Moenkopi Formation in Sinbad Valley

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(Figures 3, 4, and 5). Two of three contacts between the members correspond to polarity boundaries and the third (Tenderfoot-Alibaba contact) is within 5 m of a poorly defined magnetozone boundary zone that is marked by mixed polarities and intermediate directions. Also, the polarity boundary between the first normally magnetized and second reversely magnetized zone in the Sewemup Member (between samples 217.1 and 218) in Sinbad Valley (see Figures 3, and 4) occurs at an abrupt change in color of the strata--from red-brown below to red-orange above.

Members which are composed largely of a single rock type (such as the Tenderfoot Member, consisting of muddy sandstone, and the Sewemup Member, consisting of mudstone and shale) have relatively few polarity zones. In contrast, the Alibaba and Pariott members, each composed of interbedded coarse- to fine-grained sandstone, siltstone, and shale, possess many more magnetozones. The Alibaba, in particular, is composed of many thin polarity zones, some of which change position and/or polarity with level of demagnetization. In addition, the Alibaba shows lateral variation in polarity zonation throughout the member over distances as small as 1.6 km (see Figure 7).

One possible explanation of the correlation of polarity reversals with lithologic contacts is that changes in lithology can, in some cases, be expected to correspond to sizeable gaps in time. This line of reasoning, however, appears inadequate to explain the systematic variation in polarity reversals with lithology that occurs throughout the sequence and it leaves unexplained the correlation of the complexity of polarity zonation with member lithologic character.

The close correlation of the character of polarity zonation--both as to the complexity of the zonation as well as the position of the polarity reversals--suggests that lithology exerts significant control over the observed magnetostratigraphy.

Remanence Directions

After demagnetization at 550°C the mean direction for all reversely magnetized samples of the Moenkopi at Sinbad Valley is D = 157.2°, I = -10.2°; the corresponding mean for all normally magnetized samples is D = -14.7°, I = 17.5°. The mean directions, after demagnetization of the samples to 630°C, are essentially unchanged. The mean of the normally magnetized samples, therefore, is not antiparallel to that of the reversely magnetized samples, a result also noted for the Dolores River section of the Moenkopi [Helsley, 1969; Helsley and Steiner, 1974].

If one considers the mean directions of reversely and normally magnetized samples of each member separately, the results are similar, although in one or two cases there are so few data points that the means probably are not meaningful (Figure 10).

The simplest explanation of the lack of antiparallelism between normal and reversed means is that most samples, normal and reversed, possess a small nonremovable component parallel to the modern normal field.

This explanation is supported by the fact that most samples of all four members lose a component, during heating to 550°C, that is generally steeply downward in the northern hemisphere. This direction can be most readily explained as a mixture of components--a major one parallel to the modern field and a minor one parallel either to the normal or reversed Triassic direction. Therefore, it appears, from the nonantiparallelism of the mean Triassic directions that the modern normal component is largely but incompletely removed by thermal demagnetization, regardless of the level of demagnetization.

If one considers the means of the normal or reversed samples from each of the members separately (Figure 10), it is obvious that there is no consistently progressive change in the mean direction with stratigraphic position.

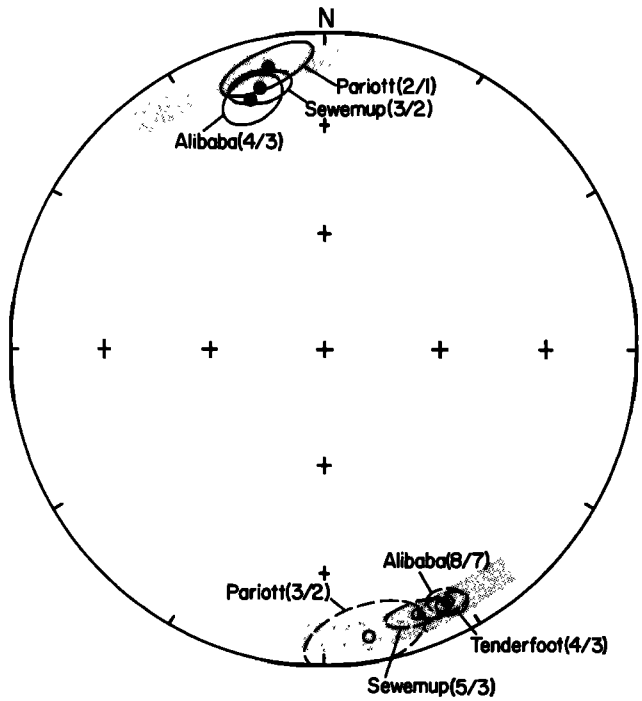


Fig. 11. Mean directions (small circles) and ellipticities at the 95% confidence level (ellipses) for the normal and reversed vectors of each member of the Moenkopi, Sinbad Valley, after cleaning at 550°C, calculated according to the method of Allridge [1974]. Normal samples in the Tenderfoot were insufficient for statistical calculation. Ellipticity (long axis/short axis) of the distribution is given by the numerical ratio. Equal-area projection: solid circles and solid line ellipses, downward; open circles and dashed line ellipses, upward. Compare with the means shown in Figure 10, calculated on the basis of Fisher statistics. Shaded arcs represent expected zones of directional shift that would have accompanied continental drift from Permian to the Jurassic. Note that the long axes of the ellipses generally coincide with and are parallel to the trend of the drift.

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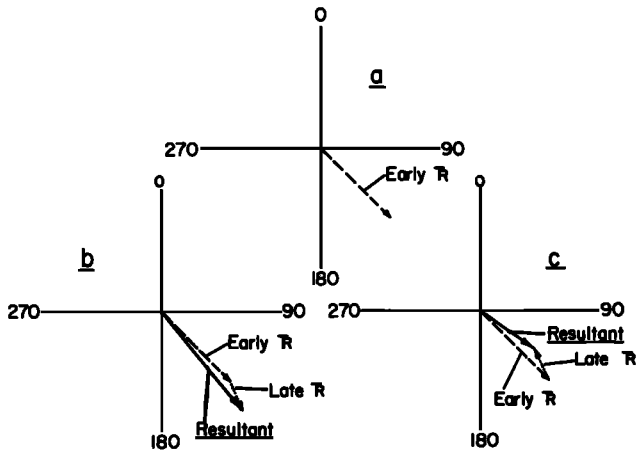


Fig. 12. Addition of latest Triassic-earliest Jurassic remanence components, either reversed (Figure 12b) or normal (Figure 12c) to early Triassic remanence (Figure 12a) to produce angular deviation in resultant remanence directions, corresponding to that from continental drift extending from about Permian through latest Triassic-earliest Jurassic. Analogous addition of normal or reversed components to an Early Triassic normal remanence direction would produce analogous results in the northwest quadrant.

Furthermore, Purucker et al., [1980] report that the mean remanence directions of all members in the stratigraphically higher Moenkopi section at Gray Mountain, Arizona are statistically the same as those reported from the Dolores River section [Helsley and Steiner, 1974]. This lack of systematic change could be interpreted to mean that either (1) little continental drift occurred in the time interval during which the Moenkopi was deposited, as suggested by Helsley and Steiner [1974], or (2) the entire sequence acquired its remanence over much the same long time interval, such that the average position of the diverse components, both normal and reversed, acquired while the continent was moving, is about the same in each member. If the latter explanation is valid, one would expect that the distribution of remanence directions of each member should show a streaked pattern corresponding to the path of continental drift that occurred during the interval of remanence acquisition. If on the other hand, the former explanation is valid, the distribution should be radially symmetric about the mean.

To test this idea, we analyzed the directional data for ellipticity, using a method described by Alldridge [1974]. Because of the difference in azimuths of the normal and reversed means, we analyzed the directional data from the normally and reversely magnetized samples separately. We restricted our analysis to those directions within 30° of the mean (to avoid biasing of results by inclusion of greatly divergent intermediate directions).

The results indicate that the directions from each member, at both the 550°C (Figure 11) and 630°C demagnetization levels are elliptically distributed. For the 550°C cleaning level, the ellipticities (long axis/short axis) for the reverse directions of the Tenderfoot, Alibaba, Sewemup, and Pariott members are 4/3, 8/7, 5/3, and 3/2, respectively, and those for the normal directions of the Alibaba, Sewemup, and Pariott members are 4/3, 3/2, and 2/1, respectively. One can conclude from these data that Fisher statistics are inappropriate for analysis of samples from the Moenkopi Formation of Sinbad Valley.

In Figure 11 we have also shown the trend of directional changes that would have resulted from continental drift during the interval from Permian to Jurassic. The major axes of the ellipses in general parallel the continental drift directional trend. This evidence supports the hypothesis that remanence acquisition occurred post-depositionally, over a long time span during which continental drift occurred.

Paleomagnetic directional data from the Dolores River section of the Moenkopi Formation also exhibit a streaking similar to that just described [Helsley and Steiner, 1974; Baag and Helsley, 1974]. In their detailed analysis of directional data from the Sewemup Member, Helsley and Steiner [1974] demonstrated the existence of an oscillatory pattern. They calculated virtual pole positions for different portions of the section and found that the poles fell on that part of the accepted polar wander path that extends in age from Permian to Jurassic [see Helsley and Steiner, 1974, Figure 7, p. 463]. They concluded that the oscillation in directions was the result of an unusual type of secular variation that tracked along the polar wander path and which was peculiar to the Triassic geomagnetic field. It appears highly unlikely to us, however, that secular variation would repeatedly conform to a continental drift path over long periods of time. A simpler explanation is that components of CRM were added over a interval lasting from a time relatively soon after deposition through the latest Triassic to earliest Jurassic. The added components could have been normal or reversed, according to the field polarity at the time of their addition. Vector addition of multiple normal and reversed components, which because of continental drift are not exactly collinear, would tend to broaden the streaking of directions (see Figure 12) over an arc corresponding to that apparently from late Permian through the latest Triassic-earliest Jurassic.

Another way to study the distribution of remanence directions in the Moenkopi is to plot a rose diagram of the declinations (Figure 13). Of the reversely magnetized samples (550°C level of demagnetization), about 75% plot in the three 15° intervals between 135° and 180°, with the maximum number (35%) plotting in the middle 15° segment (Figure 13a). In contrast, most normal directions (64%) plot in the 30° segment between 330° and 360° with the largest number (35%) falling in the 15° segment between 345° and 360°.

The nonantiparallelism of the mean directions of the normally and reversely magnetized samples described previously is also evident in this type of graphic representation. The rose diagram of azimuthal directions for samples de-

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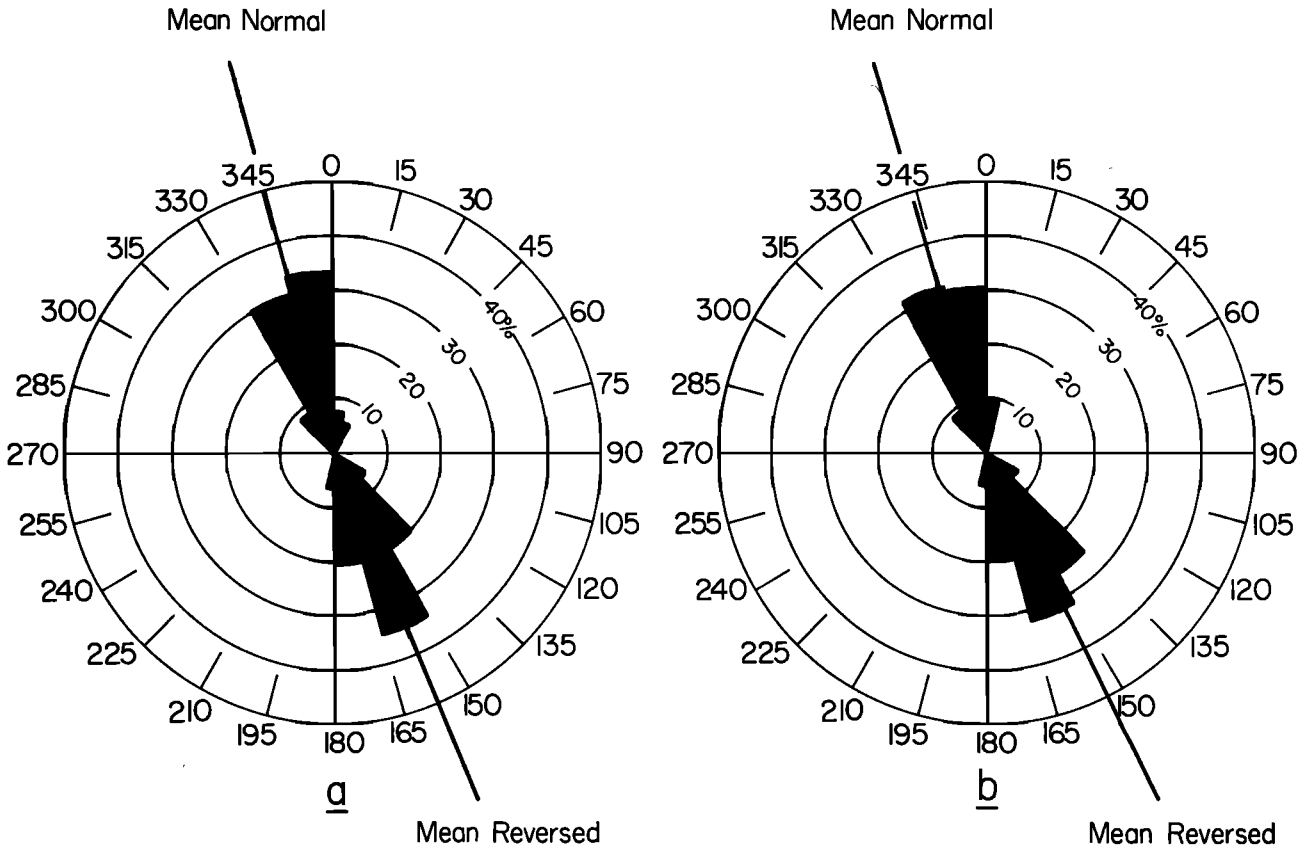


Fig. 13. Rose diagram (in 15° intervals) of the azimuths of samples from sections 1A and 1B, Sinbad Valley. (a) After demagnetization at 550°C. (b) After demagnetization at 650°C. The 15° arc segment used was arbitrarily chosen and data were shown in a segment only if they exceeded 4%. Note that normal and reversed mean azimuths in both diagrams are not antiparallel.

magnetized to 650°C displays the same lack of antiparallelism (Figure 13b).

To see if the results from any one member in any way biased the plot of the total directions given in Figure 13a above, rose diagrams of declinations (after demagnetization at 550°C) were drawn for each of the four members separately (Figure 14). With the exception of the plot of reversed directions for the Pariott Member, the normal or reversed plot of each member is similar to the combined normal or reversed plots, respectively. The number of data points for reversed samples of the Pariott Member amounts to only 20 out of a total of 172 and likely represents an inadequate statistical sampling.

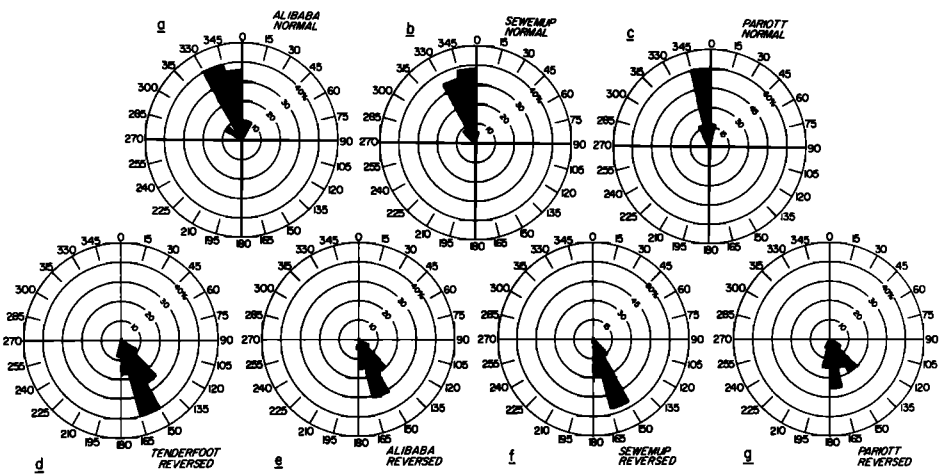


Fig. 14. Rose diagrams of the normally and reversely magnetized samples of each of the members of the Moenkopi Formation, sites 1A and 1B, Sinbad Valley, after demagnetization at 550°C.

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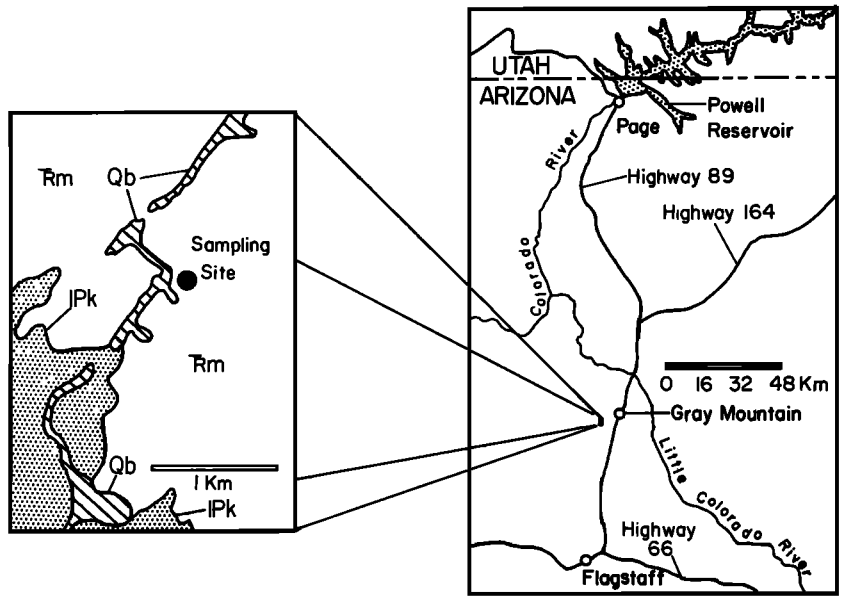


Fig. 15. Simplified geologic map of the site at Gray Mountain, Arizona; IPk, Kaibab Limestone; Rm, Moenkopi Formation; Qb, quaternary basalt.

In the rose diagram plot for the entire Moenkopi (Figure 13) it is evident that about 20% of all declinations are intermediate in direction, that is, they are more than 30° removed (to either lower or higher declination angles) from the normal and reverse means. A like occurrence of intermediate directions is common to most paleomagnetic studies of red beds [see e.g., Helsley, 1969; Helsley and Steiner, 1974; Purucker et al., 1980]. The mean log-normal intensity of samples with intermediate directions in the Sinbad Valley section is about 40% of that of the well-grouped samples. This fact indicates that the intermediate directions are not due to strong spurious IRM components such as those that might be acquired during lightning discharge. During progressive thermal demagnetization many of the samples displaying intermediate directions lost Triassic normal or reversed components. In some cases the same sample lost a component of one polarity at one demagnetization level and one of the opposite polarity at a later demagnetization level. It appears likely, therefore, that many of the weak intermediate remanence directions are the result of the presence in the same sample of nearly balanced Triassic normal and reversed components.

Intensity Data

The mean log-normal intensity of all samples of the Moenkopi Formation at Sinbad Valley after demagnetization at 550°C is 3.4×10^{-3} A m^{-1}. There is, however, considerable variation among the members. The mean log-normal sample intensities for the Tenderfoot, Alibaba, Sewemup and Pariott members are 2.3, 3.7, 4.7, and 1.9×10^{-3} A m^{-1}, respectively.

The absolute mean remanence intensities are progressively reduced during demagnetization to 630° and 650°C, but the relative differences in mean member intensities still hold.

The above data indicate the existence of a correlation between remanence intensity and lithology. The most intensely magnetized member, the Sewemup, is also the finest-grained unit; it is composed largely of mudstone, shale, and siltstone. The other three members, which are less strongly magnetized, are relatively much coarser grained. Helsley and Steiner [1974] also pointed out that coarser-grained Moenkopi beds generally tend to be less strongly magnetized than are finer-grained strata.

The data also show that remanence intensities vary considerably from sample to sample within the same member. For instance, within the Sewemup Member it is not unusual to find, after demagnetization at 550°C, some samples with intensities as high as 4.5×10^{-2} A m^{-1} and some with intensities as low as 5×10^{-4} A m^{-1}, a difference of about 2 orders of magnitude. Yet, in most cases, the lithology of the sample with high intensity appears to be essentially the same as that with low intensity. In some cases, high- and low-intensity samples are stratigraphically near one another. For example, samples 230.5 and 233.2 are within 3 m of each other and the remanence of the former is 5 times as strong as that of the latter; the remanence of sample 250.3 is about 10 times less than that of the samples about 1.5 m stratigraphically on either side of it; and sample 277.4 is about 80 times as strong in intensity as sample 280.5, some 3 m stratigraphically above it.

Fluctuation in field intensity is unlikely to be the cause of such extreme variations in sample remanence strength because in those cases in which two stratigraphically closely adjacent samples exhibit large intensity differences, the difference in time of deposition (based on stratigraphic continuity) appears to be extremely small. Petrographic evidence indicates that the hematite content for different

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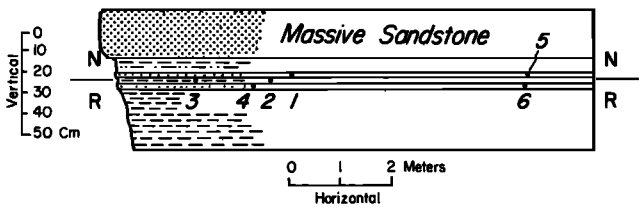


Fig. 16. Stratigraphic section of Moenkopi Formation sampled at locality A, Gray Mountain, Arizona, showing anticipated N/R magnetic boundary and location of six samples taken in the vicinity of the boundary.

samples in the Sewemup Member is essentially the same; therefore differences in hematite content within the various samples can also be ruled out as the prime cause of the large differences in intensities. Perhaps the most likely explanation of the observed intensity variation is that the remanence is composed of several components. If most components point in the same general direction, the remanence will be high; if the number of components in one direction only slightly exceeds the number of those in the opposite direction, the remanence intensity will be minimal. This line of reasoning is in accord with results from the step-wise thermal demagnetization experiments (discussed previously) which clearly demonstrated the existence of antipodal components in many samples.

Intensity data of a different type are available from study of samples from one thin stratigraphic interval within the Alibaba Member that contains authigenic cubes of hematite [see Walker et. al., 1981; Larson, 1981] which, in turn, have originated by replacement of authigenic pyrite crystals. The pseudomorphs are abundant and vary in size from 1 to 7 mm across the cube faces. Polished-section observations reveal that presently the cubes contain no pyrite relicts: All are entirely replaced by an interwoven network of specularite crystals and/or colliform hematite bands. Two stages of authigenesis, therefore are indicated--sulfidization, during the pyrite-cube growth stage, followed by oxidation during the replacement by hematite. We subjected some samples containing the pseudomorphs to selective destructive demagnetization [Larson, 1981], a technique which enabled assessment of the direction and intensity of selected large grains within a sample. The results (given by Larson [1981]) demonstrate that the remanence in the cubes is similar in direction to that in the enclosing matrix and is on the average, about 10 times stronger. A reasonable conclusion we draw from these results is that the remanence in the sandstone bed was acquired after formation and subsequent oxidation of the pyrite, long after deposition.

Gray Mountain, Arizona

The Moenkopi Formation in the Gray Mountain area has been the object of several studies involving paleomagnetism, rock magnetism, and petrography [Shoemaker and Purucker, 1974; Larson et. al., 1976; Elston and Purucker, 1979; Purucker et. al., 1980; Bressler and Elston, 1980; Walker et. al., 1981]. As previously noted, Stewart et. al. [1972] concluded that the Moenkopi section at Gray Mountain occurs stratigraphically above the sequence exposed in western Colorado along the Dolores River and in Sinbad Valley; therefore, no direct comparison of paleomagnetic results from the two areas can be made.

On the basis of an extensive paleomagnetic study, Purucker et. al. [1980] identified five polarity units in the Gray Mountain section, which they consider to be valid magnetostratigraphic zones representing remanence acquired close to the time of deposition. They noted, however, that in many portions of the various magnetozones, secondary remanence components apparently related to postdepositional in situ hematite growth could be identified. The secondary components were acquired both during the recent past and the Triassic. The modern component was considered to have resulted from near-surface weathering, whereas the Triassic overprinting was thought to have resulted from the development of hematite pigment.

Triassic overprinting of normal polarity was found to be particularly prevalent in the upper portion of the first reversed magnetozone (the Gray Mountain magnetozone). We have examined the upper boundary of this magnetozone at two separate localities in the Gray Mountain area

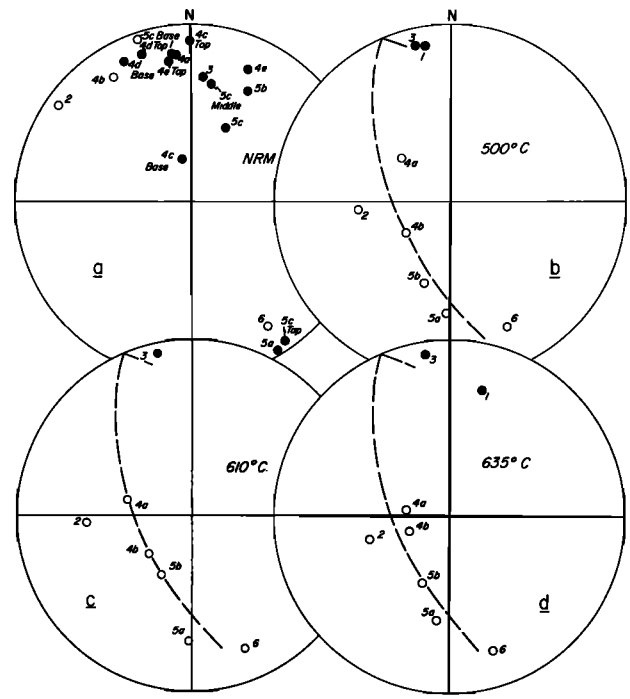


Fig. 17. Remanence directions of the samples taken at locality A initially (Figure 17a) and after thermal demagnetization steps at 550°C (Figure 17b), 610°C (Figure 17c), and 635°C (Figure 17d). Equal-area net: solid circles, downward; open circles, upward. Dashed line represents trace of great circle path along which most directions appear to lie. Note that during progressive demagnetization sample directions shift along this great circle path.

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(Figure 15) to try to determine the nature of the overprinting: the results are given below.

Locality A. At locality A (Figure 16) the boundary between the reversed zone and overlying normal zone was reported to occur abruptly within a stratigraphic interval of only 2.5 cm (E.M. Shoemaker, oral communication, 1978). Since the strata just above and below the reversal are identifiable, we could, with absolute certainty, trace it over a lateral distance of 6 m, and we collected six hand samples within that interval at or just above or below the reported polarity changeover (Figure 16). Several cores were drilled from each sample and some core specimens were cut into two or three segments. The natural remanences of these specimens are shown in Figure 17a. Samples 1, 2, 3, and 4 contain a Triassic normal direction; sample 6 is reversed; and some of the cores from hand sample 5 are normal and some are reversed. If one were to draw the polarity boundary on Figure 16 from the NRM data, it would be below sample 4 on the left, above sample 5 on the right, and it would cross the bedding planes (local time lines).

After subsequent thermal demagnetization of selected samples at 500°C, 610°C, and 635°C, the N and R groupings became more scattered (Figures 17b, 17c, and 17d). Samples 1 and 3 remained normal, and sample 6 remained reversed, but sample 2, and specimens of samples 4 and 5 moved to intermediate directions. After demagnetization at temperatures of 500°C and above, the polarity boundary would be drawn below sample 3 on the left and above sample 5 on the right of Figure 16, and therefore it would again cross time lines.

Examination of the directions after the successive thermal demagnetization steps indicates that the intermediate directions are not randomly dispersed--they all fall near a great circle that passes through the normal and reversed points (Figures 17b, 17c, and 17d).

Most of the samples are composed of muddy siltstone to fine-grained sandstone. Although each sample is generally red, the coloration is not uniformly distributed; finer-grained layers tend to be redder than coarser-grained ones. To determine if the remanence was in any way controlled by lithology, we selected a set of samples which had been thermally demagnetized at 500°C, and we sanded off as much of the redder, finer-grained zones as possible (Figure 18). In almost every case the direction moved along the previously described great circle path toward the normal direction (Figure 19). In essentially every case the movement toward the normal direction was associated with removal of a vector with a reversed Triassic direction. The conclusion drawn from these results is that these specimens contain two nearly antiparallel components of remanence which are lithologically controlled.

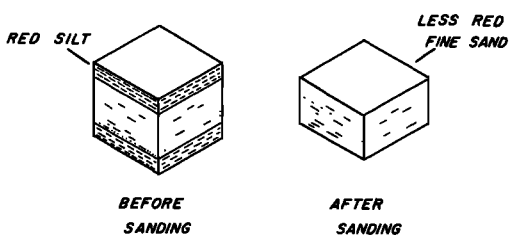


Fig. 18. Removal of red silty material by sanding.

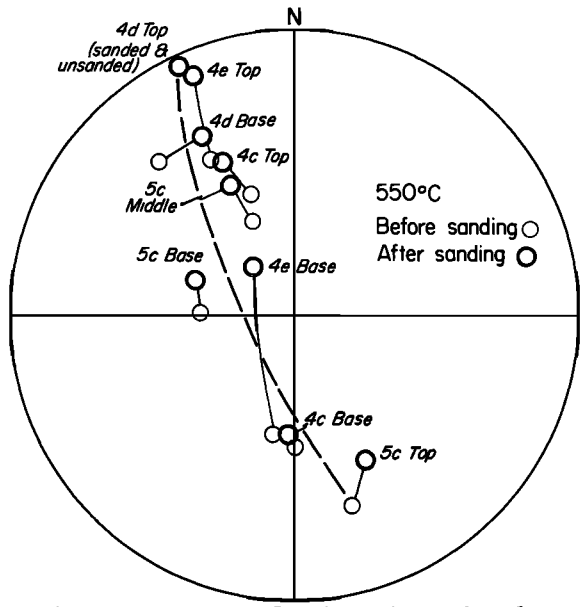


Fig. 19. Directions of selected samples from locality A, after demagnetization at 550°C, before and after removal of red silty layers by sanding. Equal-area net: open circles, upward directed. Note that sanding results in northward displacement of most remanence directions along the great circle path, indicating that the fine sand and the red silty material are of normal and reversed polarity, respectively.

The variation in directions at locality A can be modeled by the hypothetical mixing of two nearly antiparallel components in various proportions: one at declination (D) = -15°, inclination (I) = 10° downward and one at D = 170°, I = 15° upward (Figure 20). As the proportion of the two components is progressively varied, there is little shift in direction (either N or R) until the proportion is 60/40 (40/60); it is not until the ratio approaches 50/50 that appreciable shifts in direction occur along the great circle path. The intensity of the vector sum reaches a minimum at a ratio of 50/50.

This simple model provides an explanation of how a sample can contain multiple components of magnetization, yet exhibit predominantly normal or predominantly reversed directions. As long as one component exceeds the nearly antiparallel one by as little as 20%, the rock will appear to be of one polarity only. However, where the two components are nearly equal in intensity (as at locality A), intermediate directions of low intensity will result. Weak intermediate directions near a polarity boundary have commonly been considered to be evidence of a polarity transition [Baag and Helsley, 1974; Shoemaker and Purucker, 1974]. However, as the above model demonstrates, weak intermediate directions can also result from the presence of nearly equal proportions of multiple components.

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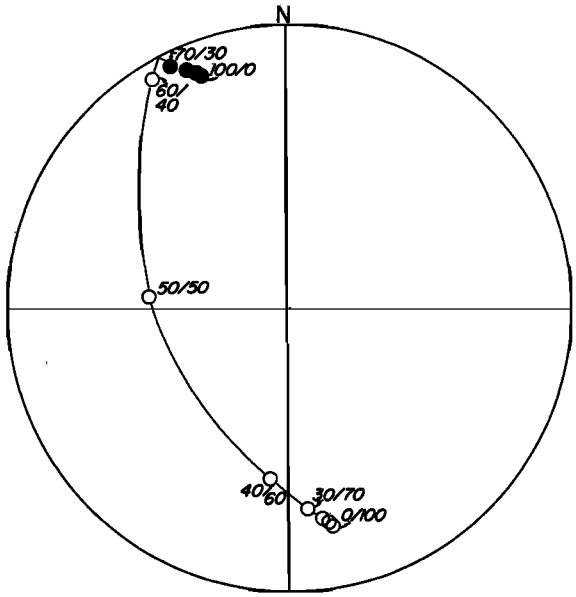


Fig. 20. Modeling of the mixing of two nearly antipodal vector components; one at D = -15°, I = 10° (downward), and one at D = 170°, I = -15° (upward). Equal-area net: solid circles, downward; open circles, upward; arcuate line is great circle path passing through the two end points and intermediate vectors. Pairs of numbers represent mixtures of components and add up to 100%: the percentage of the normal component is given by the number to the left of the slash and that of the reverse component by the number to the right. Note that for mixtures of 60% or more of one component and 40% or less of the opposite one, the directions lie relatively close to the end point normal or reversed directions. Only when the mixture gets close to 50/50 does the direction exhibit an intermediate direction.

Locality B. At locality B (Figure 21), the magnetozone boundary between reversed polarity below and normal polarity above had been previously determined to be gradational (E.M. Shoemaker, oral communication, 1978). The section at this site in upward stratigraphic succession is composed of a sequence of bottomset, foreset, and topset beds, overlain by the Lower Massive sandstone (Fig. 21).

We sampled along one foreset bed, through the topset beds and into the base of the overlying Massive Sandstone (Figure 21). Samples 1-8 all came from the same foreset bed. NRM directions of the samples are streaked out through the modern field and generally range between a Triassic normal and Triassic reversed polarity (Figure 22a). At 550°C, the directions group generally into one of two clusters, one normal and one reversed (Figure 22b). However, samples 1, 6, 7 and 8 indicate a streaking towards the normal direction reminiscent of the great circle two-component streaking at locality A (compare Figure 22b with Figure 19). Moreover, in two cases, segments cut from the same core specimen show antipodal directions: 7a and 10a are reversed, or nearly so, while 7b and 10b are normal.

Demagnetization at 615°C brings the direction of sample 8 toward the normal end point and the direction of sample 9 toward the reversed position (Figure 22c). After demagnetization at 630°C (Figure 22d), the directions of samples 8 and 2 become reversed and that of sample 10 changes back to a normal polarity.

The results of the two rock-magnetic studies (localities A and B) corroborate those of Purucker et al. [1980] in demonstrating the existence of strong, secondarily acquired Triassic components in the upper part of the first reversed magnetozone. However, our results do not support their contention that the secondary remanence resides in hematite pigment. The overprints are resistant to thermal demagnetization in excess of 630°C, indicating that the remanence in part resides in relatively large grains. Moreover, the results from the study at locality A demonstrates that the remanence polarities were lithologically dependent.

Although recognizing the presence of secondary Triassic components, Purucker et al. [1980] concluded that the characteristic remanence throughout the Moenkopi in the Gray Mountain area had been acquired penecontemporaneously with deposition. They based their conclusion on the fact that the magnetozones were well delineated (<4% anomalous directions) and areally extensive, the first reversed magnetozone, for example, being traceable across a distance of 400 km. Both of these points require comment.

1. Purucker et al. [1980] do not define what they consider to be an anomalous direction, but it is clear from their equal-area plots of remanence directions (their Figure 3) that this designation is used to describe directions more than 90° radially divergent from the mean direction. That is, any direction that plots within the hemisphere that centers on the mean direction is not considered anomalous. This criterion appears to be especially nondiagnostic. In our treatment of the data of the Moenkopi at Sinbad Valley and Maverick Ridge we designated only those directions within 30° of the normal or reversed mean directions as normal or reversed, respectively. Those beyond 30° were considered to be

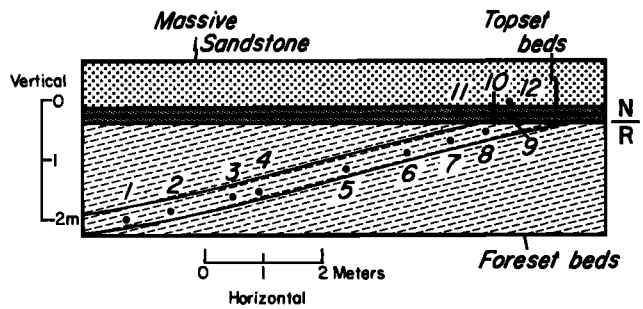


Fig. 21. Stratigraphic section of the Moenkopi Formation at locality B, Gray Mountain, Arizona, showing sampled locations. Note that samples 1-8 all came from the same foreset bed and should be all of one particular age and of one remanence direction.

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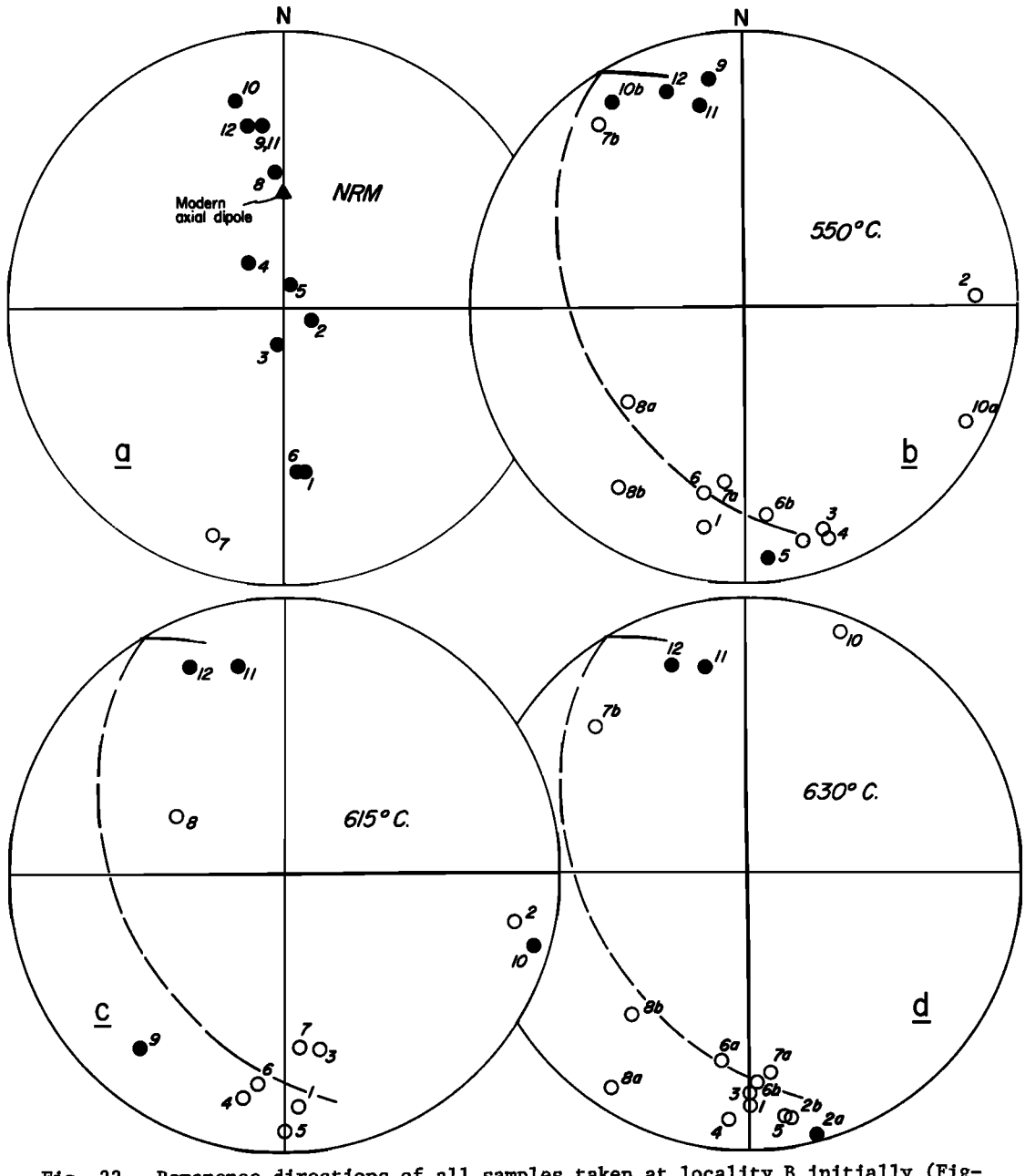


Fig. 22. Remanence directions of all samples taken at locality B initially (Figure 22a) and after thermal demagnetization at 550°C (Figure 22b), 615°C (Figure 22c), and 630°C (Figure 22d). Equal-area net: solid circles, downward; open circles, upward. Dashed line represents great circle path along which most sample directions seem to lie.

intermediate in direction. If this criterion is applied to the data given by Purucker et al. [1980] in their Figure 3, the number of intermediate directions would increase to between 15 and 25% for most magnetozones, values which are similar to those given earlier for the section at Sinbad Valley.

2. Throughout the 400 km over which the lower reversed magnetozone has been traced, it is overlain by the lower massive sandstone of McKee [1954], and throughout that distance the top of the reversed zone is restricted to a stratigraphic interval of only 2 m. In some places the top lies at the base of the overlying sandstone, in others, it is just below or above the sandstone [Purucker et al., 1980].

Similarly, the upper contact of the upper reversed magnetozone occurs within 1 m of the base of the Holbrook Member [Purucker et al., 1980]. We believe that rather than unequivocally demonstrating penecontemporaneous acquisition of remanence, this type of lithologic/magnetostratigraphic correlation could just as well be used as evidence in support of lithologically controlled secondary remanence acquisition.

Maverick Ridge, Utah

Two portions of the Moenkopi Formation were chosen for study in the Maverick Ridge area

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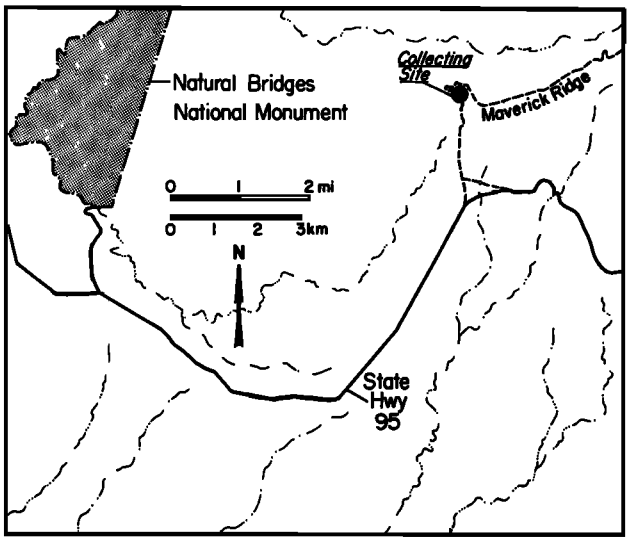


Fig. 23. Sketch map of the area in and around Maverick Ridge, Utah, showing location of sampling sites.

(see Figure 1): a massive fine-grained red to pink sandstone, and a mudstone-clast conglomerate. Both were sampled at the west end of Maverick Ridge, about 6 km east of Natural Bridges National Monument (Figure 23). Samples taken by Lienert and Helsley [1980] for their paleomagnetic study also came from this general area. Both of our sites are in the Ledge Forming Member of the Moenkopi (see Stewart et al. [1972], for stratigraphic nomenclature).

Red to pink sandstone. The sandstone, all one depositional unit, varies from red (in some places, very dark red) to pink. Several core samples were taken from both the red and pink portions and after NRM measurement they were subjected to step-wise thermal demagnetization. Petrographic analysis of polished thin sections reveals that the red samples, in contrast to the pink samples, contain relatively large amounts of authigenic hematite in the form of coatings on the silicate framework grains. Results are given for each step of demagnetization in both graphic (Figure 24) and tabular form (Table 1). In the table both the remanence directions after each heating step and the directions that were lost during each step are given. The notation used in this table is the same as that described in Figure 6. In this case, however, the mean of the closely grouped reversed Triassic directions of three red samples (RC 4, 5, and 6) after demagnetization at 580°C was taken as the direction to which all of the other directions are compared.

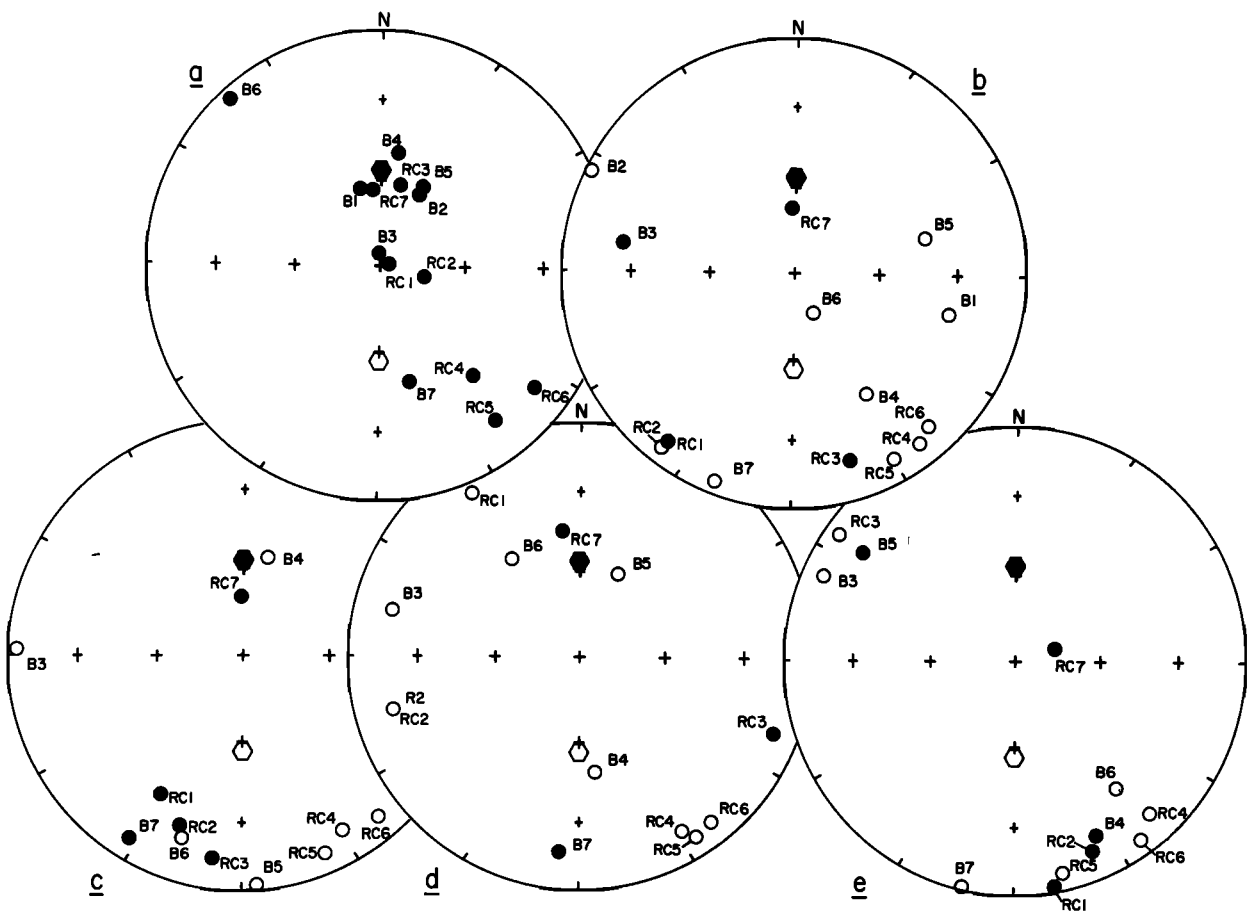


Fig. 24. Directions of magnetization of the red (denoted by RC) and pink (denoted by B) samples taken from the sandstone bed at Maverick Ridge, initially (NRM) (Figure 24a), and after progressive demagnetization at 580°C (Figure 24b), 625°C (Figure 24d), and 655°C (Figure 24e). Equal-area net: solid symbols, downward directed; open symbols, upward directed. Hexagon represents modern axial dipole direction.

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TABLE 1. Polarities of Subtracted Vectors and Samples, After Thermal Cleaning, for Maverick Ridge

Sample*	NRM**	Vector NRM-580	580°C	Vector 580-625	625°C	Vector 625-650	650°C	Vector 650-655	655°C
B1	P		IR						
B2	P		N						
B3	I		N		IN	IR	IN	I	IN
B4	P	IN(15)	R		I	IN	IR	I	R
B5	P		I	IN(12)	IR	IR	I	I	N
B6	N	N	I		IR	IR	IN	IN	R
B7	IR		IR		I	I	IR	I	IR
RC1	I		I		I	R	N	N	R
RC2	I		I		I	I	I	N	R
RC3	-		IR		I	I	R	R	N
RC4	IR		R		R		R	I	R
RC5	R		R		R		R	I	R
RC6	IR		R		R		R	I	R
RC7	P		P		P		P	N	I

*B and RC signify pink and red samples, respectively.
**P designates present-day polarity.

The NRM directions are scattered but all are in the lower hemisphere and many lie close to the modern field direction (Figure 24a). One pink sample (B6) possesses a Triassic normal direction whereas three red samples (RC4, 5, 6) are reversely magnetized in a Triassic direction, or close to it.

Demagnetization at 580°C brings about some changes in sample directions (Figure 24b). Sample B6 lost a normal component and moved to an intermediate (IR/IN) direction, whereas samples B2, B4 and RC3 (all of which possessed what appear to be modern normal NRM directions) moved to Triassic directions, B2 becoming normal and both B4 and RC3 becoming reversed. After demagnetization at 625°C most sample directions moved into the southern hemisphere, although only those of samples RC4, RC5, and RC6 can be considered good reversed directions (Figure 24c).

The distribution of directions after demagnetization at 650°C (Figure 24d) is much different from that at 625°C (Figure 24c). During this heating step, RC1 lost a reversed component and moved to a normal direction, whereas B3 and B6 both lost near-reversed vectors and moved to near-normal directions. Only RC4, RC5, and RC6 retained good reversed directions.

After treatment at 655°C, the directions of all but one sample cluster into either of two fairly tight groups, both Triassic in direction--one normal and one reversed (Figure 24e). Sample RC3 which was close to reversed in direction after heating to 650°C, became normal after treatment at 655°C through the loss of a reversed vector. On the other hand, sample B6 which was intermediate normal in direction after demagnetization at 650°, lost a component of that direction upon heating to 655°C and became reversed in direction. Samples RC1 and RC2, which were normal and intermediate, respectively, after the 650°C step, both lost normal components upon treatment at 655°C and became reversed. RC7, which had not moved significantly through heating to 650°C, finally did so upon heating to 650°C, and although the resultant direction is still intermediate, the component lost during heating was Triassic normal in direction.

It is clear from the foregoing thermal demagnetization results that most samples from the same sandstone unit at the Maverick Ridge locality contain multiple, essentially antiparallel Triassic components of magnetization, the proportion of which changes slightly during the various steps of thermal demagnetization. After the maximum demagnetization step at 655°C, some red samples are reversely magnetized and some are normally magnetized, and the same is true for the pink sandstone samples.

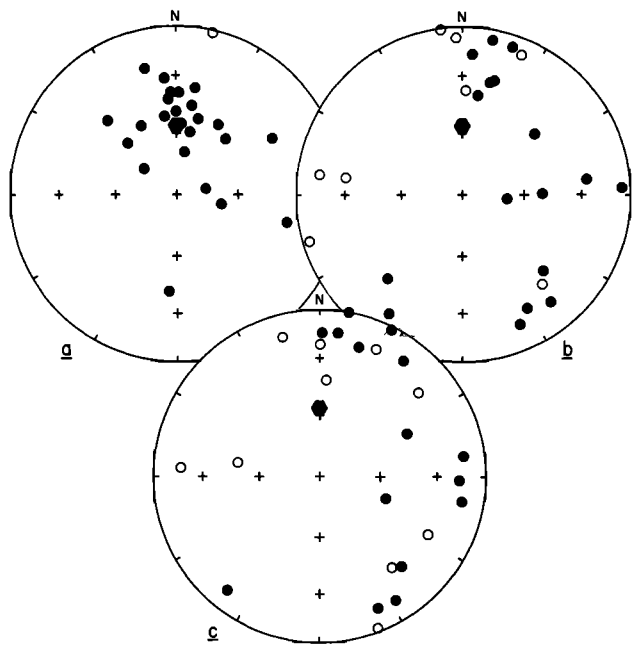


Fig. 25. Remanence directions in mudstone clasts, Maverick Ridge, Utah (Figure 25a) initially, and after demagnetization at 500°C (Figure 25b) and 625°C (Figure 25c). Equal-area projection: solid circles, downward; open circles, upward; present-day axial dipole field marked by hexagon.

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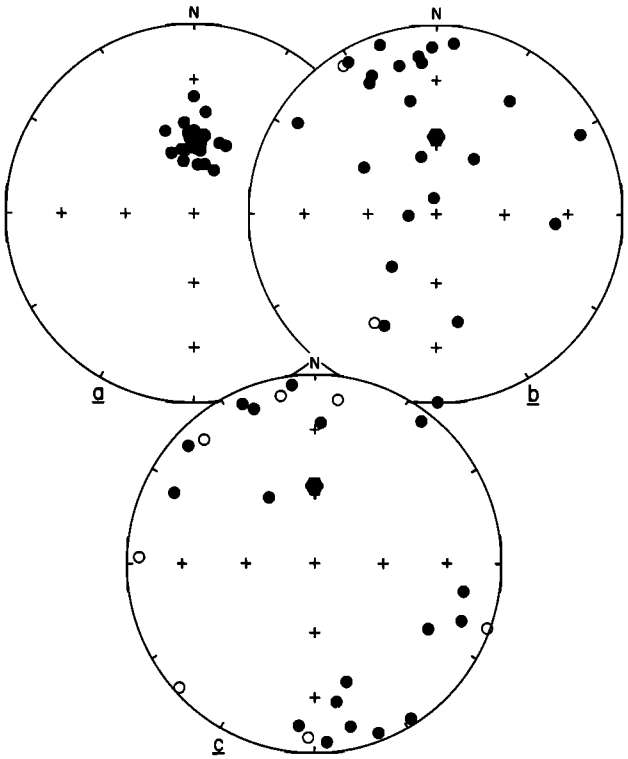


Fig. 26. Directions of vectors removed during several demagnetization intervals. (a) NRM to 500°C. (b) 580° to 610°C. (c) 625° to 635°C. Equal-area projection: solid circles, downward; open circles, upward.

Clearly, both the normal and reversed components are present simultaneously in nearly all samples, regardless of the degree of red coloration.

The above-described sandstone represents a portion of the R3 (reversed) polarity zone defined by Lienert and Helsley [1980]. In their discussion of the remanence characteristics of R3 they note that remanence directions commonly are scattered and with progressive thermal demagnetization some move toward Triassic normal directions, while others move toward Triassic reversed directions. As shown by our results, remanence directions of even the same sample can behave in the same way.

Mudstone-clast clonglomerate. Rounded cobble-size clasts of mudstone up to 30 cm across occur in the basal part of a sandstone lens in the Ledge Forming Member near the top of Maverick Ridge [McKee, 1954]. To study the nature and timing of their remanence acquisition, we obtained hand samples from 25 of the clasts and subjected them to progressive thermal demagnetization at the following steps: 400°C, 500°C, 580°C, 610°C, 625°C, 635°C, and 645°C. Randomly scattered remanence directions would be indicative of remanence acquisition prior to deposition of the clasts, whereas nonrandom groupings of directions would signify that the clasts had acquired appreciable remanence after deposition. Sample remanence directions of NRM and after demagnetization at 500°C and 625°C are given in Figure 25. NRM directions (Figure 25a) tend to be clustered around the modern axial dipole field direction, indicating the presence of sizeable recently acquired components in most samples. Directions both after demagnetization at 500°C (Figure 25b) and at 625°C (Figure 25c) shift away from the modern direction but do not become well grouped. However, the directions are not randomly distributed; they are concentrated in the eastern hemisphere and are streaked out between the average Triassic normal and reversed directions. This type of distribution is reminiscent of the great circle streaking discussed earlier for samples from Gray Mountain, Arizona. Demagnetization at 500°C dominantly removes a modern component (Figure 26a), which on the average constitutes about 75% of the mean NRM sample intensity. However,in the demagnetization interval between 580 and 610°C, about 50% of the samples lost Triassic remanence components largely in the Triassic normal direction (Figure 26b). In the demagnetization interval between 625 and 635°C about 80% of the samples lost either Triassic normal or Triassic reversed components (Figure 26c).

It seems clear, therefore, that the streaking of remanence directions which are evident in Figures 25b and 25c is the result of the presence of nearly equal antipodal remanence components--one Triassic normal, the other Triassic reversed. In their discussion of the N4 (normal) polarity unit (in which the mudstone-clast conglomerate occurs), Lienert and Helsley [1980] describe anomalous remanence directions of some samples which, during thermal demagnetization, lose a Triassic reversed component. They conclude that these samples probably possess stable secondary components.

Clearly, our results and those of Lienert and Helsley [1980] are not in accord with the assumption made by Lienert and Helsley that the acquisition of primary remanence throughout the Moenkopi Formation at Maverick Ridge (Bears Ears) was penecontemporaneous with deposition.

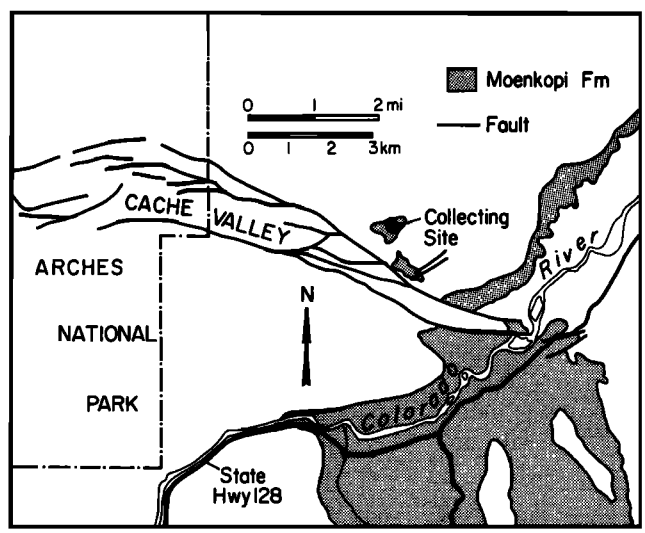


Fig. 27. Sketch map of Cache Valley, Utah, showing extent of Moenkopi Formation and location of collecting site.

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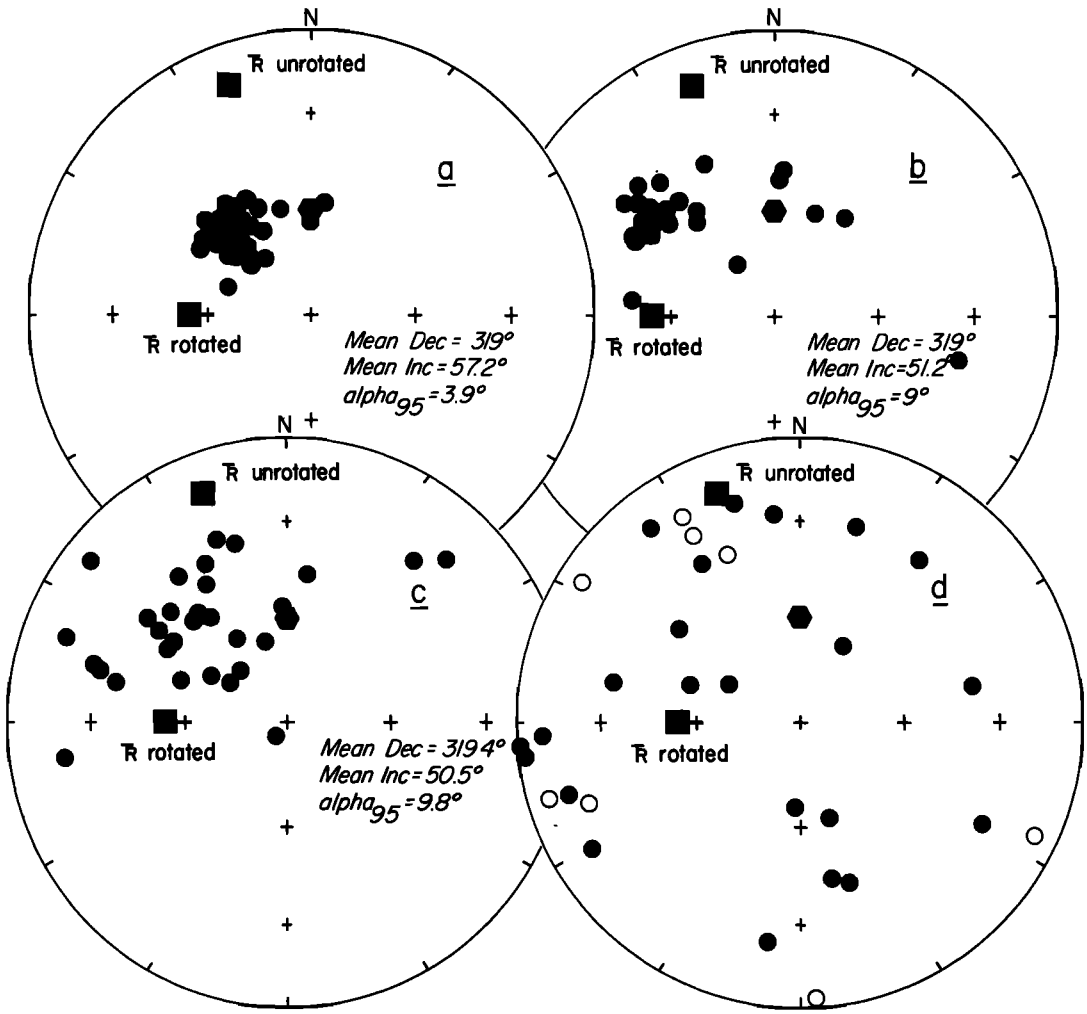


Fig. 28. Directions of remanence of samples from Cache Valley site after demagnetization. (a) 500°C. (b) 565°C. (c) 610°C. (d) 625°C. Without application of dip corrections. Equal-area net: solid symbols, downward directed. Hexagon is axial dipole direction; squares represent the Triassic normal direction, one unrotated and the other dip corrected.

Cache Valley, Utah

Cache Valley (Figures 1 and 27) is one of several northwest-southeast-trending topographic depressions in southeastern Utah that have developed in response to the localization of erosion along anticlines developed over salt diapirs. Within Cache Valley, about 4 km east of Arches National Park Headquarters, is a section of Moenkopi strata dipping about 85° to the north. We studied the remanence directions of samples taken from this section to determine whether the magnetism was acquired prior to or after the folding which can only be dated as Triassic to Cretaceous [Dane, 1935]. This locale is one of the few in which a test of this type could be applied. At most places on the Colorado Plateau the Moenkopi beds are essentially horizontal.

We took 35 samples across a stratigraphic interval of about 5 m and after NRM measurement subjected them to thermal demagnetization at 500°C, 565°C, 610°C, and 640°C. Remanence directions after demagnetization at 500°C are well grouped about a mean that does not correspond either to an unrotated Triassic normal direction or to a dip-corrected Triassic normal direction (Figure 28a). Only three of the points (without structural correction) lie close to the modern axial dipole direction. NRM directions, which are not shown, are essentially the same as those after demagnetization to 500°C.

Demagnetization at 565°C brings about an increase in scatter: About 5 directions lie close to the modern day axial dipole direction and two are intermediate. The mean direction of all samples, after demagnetization at 565°C, is D = 319.0°, I = 51.2° (Figure 28b). Again, there appears to be no correlation of this direction with either the structurally compensated or uncompensated average Triassic normal direction.

Demagnetization at 610°C brings about a significant increase in scatter, even though the mean direction of all points is similar to that after 565°C: D = 319.4°, I = 50.5° (Figure 28c), and after thermal cleaning at 640°C the direc-

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tions appear to be randomly distributed (Figure 28d).

For samples from this locality, the most meaningful grouping is isolated by demagnetization at 500° and 565°C and the corresponding mean directions, without structural dip correction, appear most similar to the average Cretaceous direction [Irving, 1964; McElhinney, 1968a, 1969]. A simple interpretation of the results from Cache Valley is that the observed well-grouped directions were acquired after folding, through authigenesis that occurred sometime during the Cretaceous. Little or no trace of any remanence that might have been acquired during the Triassic has remained.

Transmitted-light petrographic studies of polished thin sections reveals that the samples from Cache Valley are fine-grained sandstone cemented throughout by dolomite. The cement is nonuniformly stained red and pink by hematite pigment. Observations by reflected light indicate that: (1) detrital iron-titanium oxide grains are rare (of those present most consist of leucoxene, the alteration product of ilmenite; martite grains are essentially nonexistent), and (2) hematitic microcrystalline rosettes (1 to 2 μm in diameter), which appear as blood red 'beads' when viewed under vertical illumination, occur in great quantities and are restricted to the dolomitic cement and are readily apparent when samples that have been etched in HCl are examined under the scanning electron microscope.

The petrographic data are consistent with the remanence behavior during thermal demagnetization. The principal carriers of remanence are the hematite rosettes and beads, which are so fine-grained that their blocking temperatures are largely exceeded at temperatures above 600°C. Since there are few large hematite grains present, the remanence directions become progressively scattered during progressive thermal treatment.

The Moenkopi sandstone at Cache Valley appears to be a relatively normal clastic sediment and it is likely that magnetite (and/or martite) grains would have constituted a significant fraction of the accessory grains when the sand was initially deposited. The present lack of such grains has apparently been the result of intrastratal dissolution. Obviously the hematite in the dolomitic cement is secondary. From a consideration of remanence directions it appears most likely that this material was precipitated subsequent to dissolution of the detrital opaques, after folding, during the Cretaceous.

Cutler Conglomerate Test

Twenty-five samples were obtained from granitic and metamorphic cobbles and small boulders, some up to 25 cm across, in a comglomerate lens in the upper part of the Permian Cutler Formation exposed in and near a roadcut along Colorado State Highway 141 about 3 km south of Gateway, Colorado (see Figure 1).

The pebbles and cobbles were derived from the same general source as were the finer-grained arkosic sediments of the Moenkopi Formation--the Uncompahgre Uplift which remained

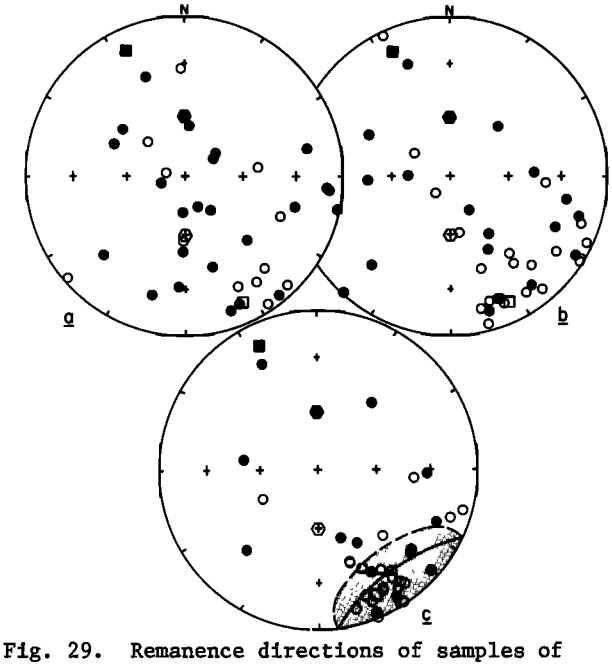


Fig. 29. Remanence directions of samples of boulders and cobbles in a Cutler Formation conglomerate lens, near Gateway, Colorado, initially (Figure 29a) and after af demagnetization at 3.2×10^4 A m^{-1} (Figure 29b) and 8×10^4 A m^{-1} (Figure 29c). Equal-area projection: solid symbols and solid lines, downward; open symbols and dashed lines, upward. Hexagons, modern axial directions; squares, mean Permo-Triassic directions, Cutler Fm, western Utah given by Gose and Helsley [1972]. Hatchured region represents cone of 30° radius around mean direction.

in existence from the Pennsylvanian through the Triassic periods and along whose crest Precambrian basement rocks were exposed. Without doubt, the remanence directions of these large clasts were randomly distributed at the time of deposition. We sampled the gravels to see if identifiable secondary remanence components had been acquired since deposition.

After NRM measurement, all were subjected to thermal demagnetization at 400°C and most of them underwent complete physical disintegration. Petrographic observation revealed that biotite, originally an abundant constituent in the samples, had altered to vermiculite. Upon heating, this material expanded and brought about the disaggregation. We sampled an additional 35 clasts and, following NRM measurement, subjected them to progressive af demagnetization to peak fields as high as 8×10^4 A m^{-1} in order to diminish the remanence carried by magnetite and to enhance relatively that carried by the higher-coercivity hematite (Figure 29).

The NRM directions appear scattered although most plot in the southeastern quadrant (Figure 29a). By 3.2×10^4 A m^{-1}, three quarters of the sample directions group in the southeastern quadrant and nearly half of all samples possess a Triassic reversed direction (Figure 29b). Further demagnetization at 8×10^4 A m^{-1} brings about further improvement in grouping (Figure 29c); over half of all sample

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directions (18/35) are reversed, lying within 30° of the mean reversed Triassic direction, and 10 others are nearly reversed. One sample direction is Triassic normal. The mean direction of all 35 samples (with the normal direction inverted) is D = 144.6°, I = 6.9°, with an α_{95} of 14.4°. This corresponds to a Permo-Triassic direction, and the grouping is as good as almost any we determined from much finer-grained red sedimemts of the Moenkopi Formation in Sinbad Valley.

For comparison, we sampled the finer-grained matrix surrounding the gravel clasts and measured its remanence after step-wise thermal demagnetization to 630°C. The mean direction of 6 samples after heating to 630°C is very close to the mean of the gravel clasts after af demagnetization to 8 x 10^4 A m^{-1}--D = 140.3°, I = -9.1°.

In addition, five samples from this conglomerate have been subjected to selective destructive demagnetization [Larson, 1981] to determine what forms of hematite carry the remanence. Under reflected light it can be seen that the gravel clasts contain three principal forms of hematite analogous to some of those reported by Walker et al. [1980]: (1) martite, representing pseudomorphic replacement of magnetite, (2) pods of precipitates and replacements in biotite grains, and (3) crack fillings and stains of ultrafine grain size throughout the rocks.

The results (reported by Larson [1981]) show that all three forms carry some of the remanence, the proportion of which varies from sample to sample, and that any one of the three forms can be the dominant component. The results also show that the direction of remanence in the martite and in crack fillings and stains is variable, in some cases being Triassic normal and in others Triassic reversed in direction. The remanence direction carried by the hematite pods in biotite grains was in all cases of reversed Triassic polarity.

Obviously, the clasts have acquired their remanence over a long time span during which authigenesis occurred, and the field apparently repeatedly changed in polarity. What we see in any one core sample is a resultant direction made up of multiple components.

The results of this conglomerate test show clearly that the clasts possess consistent remanence components that could only have been acquired after deposition. As pointed out earlier, the gravel clasts in the Cutler Formation were derived from the same source material as the finer-grained material that makes up the bulk of the Moenkopi strata. Basically the only difference is one of grain size--the overall mineralogy of the original sediments appears to have been essentially identical. Alteration of clasts of cobble and boulder size must have been a relatively slow authigenic process yet the observable magnetization of core-sized specimens is relatively consistent. If clasts of this size can acquire a stable, generally consistent remanence through addition of multiple nearly antiparallel secondary components, finer-grained material of the same composition such as that of the Moenkopi Formation could have done the same.

Petrographic Observations

Petrographic data for most of the strata discussed above have been given in detail by Walker et al. [1981]. In brief, hematite, the principal ferromagnetic material in the rocks, consists of a variety of forms of variable grain size and origin. Most of the forms originated after deposition and they commonly reflect multiple generations of diagenesis.

Model of Remanence Acquisition

The results of our study, coupled with the results of other studies reported in the literature, permit the synthesis of paleomagnetic, rock magnetic, and petrographic data into a reasonably comprehensive model of remanence acquisition in the Moenkopi Formation. The features for which a model must provide an explanation include (1) the presence of both normal and reversed components in many of the same samples, at many localities in the formation, (2) the close correlation between lithology and both remanence intensity and polarity in the Sinbad Valley section, (3) the occurrence of generally weak intermediate directions in about 20% of the Sinbad Valley samples, (4) the lack of a progressive change in mean remanence directions of the four members from the base upward in Sinbad Valley (as well as in the members of the stratigraphically overlying section, at Gray Mountain, Arizona), (5) the streaked distribution of remanence directions in any one member, and in the formation as a whole in the Sinbad Valley section--a pattern which appears to coincide with the accepted polar wander path, and (6) the preponderance of authigenic hematite, in a variety of forms, that in most cases can be shown or inferred to have formed long after deposition.

Nature of Original Sediment

We take for our initial sediments, as in the case of most of the late Paleozoic and Mesozoic red beds on the Colorado Plateau, first-cycle detritus derived from the relatively rapid erosion of nearby highlands of the so-called Ancestral Rockies of Colorado, Arizona, and New Mexico. Uplift and erosion in the source areas exposed Precambrian plutonic and high-grade metamorphic rocks. Sediments derived from these rocks should have been rich not only in quartz and feldspars but also biotite, hornblende, magnetite, and ilmenite, because these minerals are common in modern sediments derived from the same Precambrian source rocks that have been re-exposed during and since the early Tertiary Laramide orogeny.

The proportions of the various minerals would undoubtedly have varied with the dominant grain size of the particular sedimentary layer: finer-grained beds would have contained a higher proportion of biotite and clay (much of it iron-bearing), whereas coarser beds would likely have contained, proportionally, more quartz and feldspar.

Undoubtedly, the permeability of the sedimentary sequence to groundwater flow would have

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been variable owing to initial variations in the texture of the sediment.

Initial Remanence

When first deposited it is likely that the sediments would have carried some level of depositional or postdepositional remanence [Verosub, 1977] acquired during or slightly after deposition.

On the basis of the data and arguments put forth in our previous article [Walker et al., 1981] we believe it is likely that the principal ferro-magnetic detrital material derived from the source area would have been unaltered or only partly altered magnetite. Our best estimate, based on grain counts of polished thin sections of samples from Sinbad Valley and the Gray Mountain locality, is that the incoming mixture of detrital opaques would have been composed of magnetite (∿65%), ilmenite (∿20%), monocrystalline and polycrystalline hematite (∿10%), and ilmenite-hematite intergrowths (∿5%). Weathering in the source area may have led to the development of thin martite rims on some of the larger magnetite grains and perhaps even to complete martitization of some of the smaller ones. However, most magnetite grains, we envision, would have been largely unaffected by oxidation prior to deposition. Hematite, because of its comparatively low magnetic intensity and low proportion in the detrital grains, would carry only a minimal amount of the remanence. Finer-grained sediments tend to carry more of the small, magnetically stable single- and pseudosingle-domain grains of magnetite, and these tend to be deposited under lower-energy conditions than the coarser-grained sediments. Therefore, the finer-grained strata would probably have been (for the same quantity of magnetite) more strongly and coherently magnetized than the coarser-grained sediments.

Alternatively, we consider the case (which we feel is much less likely) in which oxidation in the source area and during transportation was sufficient to bring about nearly complete oxidation of the magnetite to hematite prior to the time of deposition. In this case, an initial DRM (or postdepositional remanent magnetization, PDRM) carried by hematite would have been acquired by the sediments at or soon after deposition. The magnetic moment of hematite is considerably less than that of magnetite, however, so that the resulting remanence would have been proportionally less than for the case outlined above.

Intrastratal Alteration

At some time after deposition of part or all of the stratigraphic section, subsurface conditions were favorable for intrastratal alteration. Chemical alteration is a function of the groundwater chemical environment which can vary areally, stratigraphically, and temporally. Under some conditions, such as when early cementation seals off the permeability of the sediments, even the chemically most unstable minerals, such as hornblende, may remain unaffected; whereas under other conditions, such as when deep burial increases temperature and pressure, even generally stable minerals like quartz and potassium feldspar may undergo significant alteration. Conditions affecting alteration would not have remained constant but would have varied in response to changing subsurface conditions. Reducing conditions might exist for a time, to be superseded by oxidizing conditions. The color and the preponderance of hematite in relation to other opaques indicate that conditions in red beds eventually become predominantly oxidizing.

Ions available for alteration of existing phases and growth of new ones can be supplied from a distant source and carried in through groundwater circulation or from a local source by the breakdown of pre-existing minerals. Iron ions, for example, which can be used to build authigenic ferric-iron-oxide minerals, such as hematite, can be supplied by the breakdown of iron-bearing silicates--biotite, hornblende, iron-rich clay--and/or some detrital iron-oxide grains (ilmenite, magnetite) in which some or all of the iron ions are initially in the ferrous state. Under oxidizing conditions, detrital grains of magnetite, if present, would be unstable and would begin to convert to hematite, and in the process, as a result of the loss in saturation moment, the initial remanence would be greatly decreased. A point of contention is whether the resultant remanence direction maintains the initial DRM direction or takes on that of the ambient field. Because of the change in crystal symmetry during martitization, from isometric to hexagonal, it is likely that the magnetization in the newly formed hematite will reflect the ambient field. Recent experimental work supports this conclusion [Amerigian, 1978]. Moreover, the data we have obtained from selective destructive demagnetization studies of the gravel clasts from the Cutler Formation [Larson, 1981] demonstrates that in these sediments, in the changeover from magnetite to hematite (martitization), the grains acquired consistent PermoTriassic remanence directions.

From our petrographic studies, it has also become evident that detrital ilmenite grains undergo oxidation with time, forming in some cases leucoxene and in others a blotchy two-phase mixture of hematite and a mineral which, although not definitely identified, appears to be an iron-titanium oxide [see Walker et al., 1981]. The hematite patches, which are quite abundant and range in size from 0.1 to 10 μm, are capable of contributing a significant secondary component to the total remanence. The intergrowths are the result of a slow-acting low-temperature process which requires time not only for the oxidation of the ilmenite but also for the subsequent diffusional separation of the oxidation products into mutually exclusive mineral phases. The formation of these kinds of grains almost certainly requires longer-lasting conditions than those encountered during soil-forming processes. There can be little doubt therefore that their development is the result of intrastratally induced authigenesis and, accordingly, that their remanence is a CRM developed long after deposition.

In addition to the oxidation of detrital

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opaque grains, chemical break-down of ferrous silicates can occur, producing iron ions which, under oxidizing conditions, will result in the precipitation and growth of crystals of authigenic hematite. Each new crystal will acquire a remanence reflecting the ambient field direction at the time it exceeds the critical grain size; subsequent crystal growth should stabilize and intensify the initial remanence. Diagenetic alteration will lead to (1) growth of single crystals and rosettes of hematite in interstitial voids, (2) replacement of framework iron-silicate grains by hematite intergrowths, and (3) growth of hematite pods in biotite grains; all of which are common hematite occurrences in the Moenkopi Formation [Walker et al., 1981].

It must be kept in mind that because of lithologic and textural differences throughout the stratigraphic sequence, rates and degrees of alteration will vary. While one interval reaches a certain level of secondary remanence acquisition, another may have lagged far behind, and still another will have greatly exceeded it. The length of the period of hematite authigenesis will correspond to the time period that the Eh-pH conditions are appropriate and during which unstable iron-bearing grains are undergoing alteration. Moreover, throughout the geologic history of the stratigraphic sequence, subsequent periods of alteration that release iron ions can occur whenever the proper conditions exist.

It is apparent from the common occurrence of nearly antiparallel vectors in individual Moenkopi samples and from the streaked pattern of the sample directions which tends to coincide with 30 to 45° of the polar wander path that alteration probably continued over a time span of several tens of millions of years. It is likely, however, that authigenesis (and remanence acquisition) will proceed most rapidly during the earlier stages of alteration, when there is relatively a greater proportion of unaltered iron-bearing opaque minerals and unstable iron-bearing silicate grains. Each stratigraphic interval of similar lithology will be expected to behave as a unit, generally altering at rates and to extents different from other lithologic units. Over relatively short time intervals during the peak rates, it is probable that either the normal or reversed geomagnetic polarity will dominate. Hence, it is likely that in any one lithologic unit, one polarity will dominate.

It is apparent in our model that the resulting polarity sequence is strongly controlled by lithology. Where the lithology includes a relatively large number of different rock types, such as conglomerate, sandstone, siltstone, and shale within a relatively thin stratigraphic interval, it is highly likely that a relatively large number of polarity zones will be present (as in the case of the Alibaba Member of the Moenkopi Formation in Sinbad Valley). On the other hand, where the lithology remains rather uniform, polarity zonation will generally be simple (as in the Tenderfoot and Sewemup members). Polarity switches, therefore, should tend to occur at or near changes in lithology and thus commonly coincide with member boundaries (as is commonly the case in the Moenkopi Formation). Areally extensive polarity changes would also be expected to be associated with areally extensive lithologic units as, for example, the R1/N2 and N2/R2 contacts at or near the bases of the lower massive sandstone of McKee (1954) and the Holbrook Member, respectively, in the Gray Mountain area, Arizona [Purucker et al., 1980].

In some parts of any one polarity unit, no single polarity will be dominant; rather the rocks should have acquired nearly equal antiparallel components. Samples from these zones should be marked by intermediate directions and weak intensity, characteristics quite common to the Moenkopi and other red beds as well. During the alterational history of any one polarity unit, variation in the magnitude of the normal and reversed components added throughout the time span of authigenesis can locally result in directions nearly antiparallel to those of surrounding rocks; again, a common feature in the Moenkopi.

Near polarity boundaries, where there is a changeover from one polarity to another, any one of several situations can exist depending on the local alteration history of the strata. In some cases the polarity switch may be abrupt with no transition zone; in others, there may be a zone of intermediate weak directions (where normal and reverse components nearly neutralize each other); and in others, the changeover might be marked by multiple, stratigraphically abrupt switches in polarity (such that in one sample the normal component exceeds the reverse one, in the next the opposite is true, and so on). All three of these cases are well documented in the Moenkopi Formation [our own data, Baag and Helsley, 1974; and Purucker et al., 1980].

In addition to lithologic control, of course, rates of alteration can also be controlled by other processes such as groundwater table levels, cementation, and dissolution of previously acquired cement. It is not unlikely that Eh-pH conditions will be different above and below a groundwater interface. Resultant conditions could affect the nature and extent of alteration basin-wide. Cementation, which effectively seals off portions of the sedimentary section from contact with groundwater, can effectively freeze the level of authigenesis. And, of course, subsequent dissolution of previously precipitated cement can reopen the rocks to the effect of chemically active solutions.

When one considers all of the possible situations that a stratigraphic sequence such as the Moenkopi Formation could have undergone during the many tens of millions of years of its existence, it seems plausible to us that intrastratal alteration, lithologically controlled and acting at reasonable geologic rates on first-cycle detritus, appears to be capable of explaining all of the magnetic and petrographic features found in the Moenkopi Formation, both on a small (microscopic) scale and on a large (from sample to magnetozone) scale. Moreover, this basic model for the acquisition of CRM over geologically long time intervals seems generally applicable to nearly all red beds.

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